MIL-T-55605
2 August 1968

MILITARY SPECIFICATION

TEST SET, TRANSPONDER AN/APM-123( )(V)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. - This specification covers one type of portable go-no-go test set that performs flight line operational tests of airborne Identification Friend or Foe (IFF) transponder sets. The test set is designated Test Set, Transponder AN/APM-123( )(V) and is intended for tri-service use. (See 6.1 and 6.6.)

1.2 Classification. - The Test Set, Transponder AN/APM-123( )(V) shall be of the following versions, as specified: (See 6.2.)

Version	Designation	User
1	AN/APM-123( )(V)1	US Army
2	AN/APM-123( )(V)2	US Navy
3	AN/APM-123( )(V)3	US Air Force

2. APPLICABLE DOCUMENTS

2.1 Documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder, Tin Alloy, Lead-Tin Alloy and Lead Alloy
PPP-B-20	Bag, Cotton, Mailing
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

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PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	Varnish, Moisture-and-Fungus-Resistant (For the Treatment of Communication, Electronic, and Associated Electrical Equipment)
MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-T-713	Twine, Impregnated, Lacing and Tying (For use in Electrical and Electronic Equipment)
MIL-I-3158	Insulation Tape, Electrical Glass-Fiber (Resin-Filled), and Cord, Fibrous-Glass
MIL-I-3190	Insulation Sleeving, Electrical, Flexible, Treated
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys, Spot and Seam
MIL-I-7444	Insulation Sleeving, Electrical, Flexible
MIL-I-7798	Insulation Tape, Electrical, Pressure-Sensitive Adhesive, Plastic
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, and Nickel Alloys
MIL-M-10578	Metal Conditioner and Rust Remover (Phosphoric Acid Base)
MIL-I-11748	Interference Reduction for Electrical Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish for Ground Signal Equipment
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base)
MIL-T-15659	Terminal, Lug, Solder, Copper and Phosphor Bronze
MIL-I-18057	Insulation Sleeving, Electrical, Flexible, Glass Fiber, Silicone Rubber Treated
MIL-I-21557	Insulation Sleeving, Electrical, Flexible, Glass Fiber, Vinyl Treated
MIL-I-22076	Insulation Tubing, Electrical Nonrigid, Vinyl, Very Low Temperature Grade
MIL-I-22129	Insulation Tubing, Electrical, Polytetrafluoro-ethylene Resin Nonrigid
MIL-I-23053	Insulation Tubing, Electrical, Heat-Shrinkable

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MIL-S-23190	Strap, Cable, Adjustable, Plastic
MIL-B-43291	Box, Fiberboard, Corrugated, Double-Wall, Weather Resistant
MIL-P-55110	Printed Wiring Boards
MIL-T-55156	Terminals, Lug, Splices, Conductor, Screw Type, General Specification for

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 Metals; Test Methods

MILITARY

MIL-STD-22	Welded-Joint Designs
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Requirements for Reliability Program (for Systems and Equipments)
MIL-STD-810	Environmental Test Methods
MIL-STD-1277	Splice, Terminal, Terminal Board, Binding Post, Electrical
MS 35436	Terminal, Lug, Solder Type, Copper Stamping, Insulation Grip, One Hole
MS 35440	Terminal Lug, Solder Type, Copper Stamping, Insulation Grip, Slotted

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Testers
SC-B-61578	Grounding to Chassis
SC-DL-568301	Test Set, Transponder AN/APM-123( )(V)
SM-D-568404	Test Set, Transponder AN/APM-123(V)1
SM-D-568405	Test Set, Transponder AN/APM-123(V)3
SM-D-568406	Test Set, Transponder AN/APM-123(V)2

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PUBLICATIONS

DEPARTMENT OF DEFENSE

AIMS 64-900(Classified-Confidential)	Mode 4 Input/Output Data KIR/KIT-1A/TSEC
AIMS 65-1000(Classified-Confidential)	Technical Standard for the ATCRBS/IFF/Mark XII System

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description. - The Test Set, Transponder AN/APM-123( )(V) consists of a Test Set, Transponder TS-1809( )/APM-123(V) and associated cable assemblies, an attenuator, a test hood, an altitude code conversion computer and a case with removable cover and removable shoulder strap. The TS-1809( )/APM-123(V) interrogates the transponder under test and evaluates replies to determine the adequacy of the transponder's performance. The TS-1809( )/APM-123(V) checks operational characteristics such as receiver frequency and sensitivity, decoding and coding of all modes, including Mode 4 (when using suitable external equipment), transmitter output power and frequency, and sidelobe suppression. Acceptance or rejection of any of these parameters tested is visually indicated by illumination of accept or reject indicators. The test set contains an L-band interrogator (modulated by a coder), a receiver, a decoder, and a self test feature. By means of a front panel switch, the test set is capable of operating from either an external 115 Vac or 28 Vdc power source. The accessories furnished with the AN/APM-123( )(V) permit performing umbilical tests, in addition to the free space coupling tests.

3.2 First Article - Preproduction test samples. - The contractor shall furnish three First Article - Preproduction test samples for approval if required by the invitation for bids and contract. (See 4.3 and 6.2(f).)

3.2.1 Fabrication and assembly. - First Article - Preproduction samples shall be fabricated by the contractor and shall meet all requirements specified for the equipment on contract. The First Article samples shall be assembled using parts, materials, and processes that will be employed in production, and shall be fabricated and assembled in a manner similar to that to be used in production.

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3.2.2 Approval and use. - Approval of First Article - Preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (See 3.2.3), First Article samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.2.3 Reference standards. - After First Article - Preproduction samples have been approved, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.3 Construction. - The equipment shall be constructed in accordance with the following:

Drawing and Data List		Assembly Drawing	Item Designation
SC-DL-568301	Column 1	SM-D-568404	AN/APM-123( )(V)1
SC-DL-568301	Column 2	SM-D-568406	AN/APM-123( )(V)2
SC-DL-568301	Column 3	SM-D-568405	AN/APM-123( )(V)3

3.4 Accessories. - The accessories established for the various versions of the equipment are required in order to satisfy the particular user's needs. See 6.3 for detailed listing of components to be furnished as part of the test set on order.

3.5 Cleaning. -

3.5.1 Parts. - After fabrication, parts shall be cleaned in accordance with requirement 9 of MIL-STD-454, or as specified in an applicable document.

3.5.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.6 Finish. - The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4.)

3.7 Marking. -

3.7.1 General. - Marking shall conform to Specification MIL-M-13231. (See 4.4.) Front panel marking shall be group I as described in that specification. Marking of printed circuit boards, however, and the components mounted on the boards shall be in accordance with Specification MIL-P-55110.

3.7.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

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3.7.3 Serial numbers. - The following items shall have serial numbers:

(a) Test Set, Transponder AN/APM-123( )(V) (top and front of case)
(b) Test Set, Transponder TS-1809( )/APM-123(V)

3.8 Riveting. - Riveting shall not be used for mounting parts such as capacitors, resistors, transformers, inductors, and tube sockets. Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. The distance from the centers of rivet holes to the edges of the material in which the holes are formed shall be at least 1½ times the rivet diameter. Rivets shall be used where practicable in preference to other means of securing aluminum or magnesium structural parts if the joint is subjected to dynamic tension loads when the equipment is exposed to specified service conditions and if the parts do not require removal when the equipment is serviced. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.9 Securing of parts. -

3.9.1 General. - Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the items in which they are used are subjected to specified bounce and shock tests. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements. When practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely.

3.10 Brazing and soldering. -

3.10.1 Brazing. - Brazing of steels, copper, copper alloys, aluminum, aluminum alloy, and nickel alloys shall conform to Specification MIL-B-7883.

3.10.2 Soldering. - Soldering for applications other than those using automatic flow or dip soldering for printed circuit assemblies shall conform to requirement 5 of Standard MIL-STD-454. Solid solder shall conform to type S and fluxcored solder to type R or RMA of Specification QQ-S-571. Liquid rosin flux for use with solid solder shall conform to MIL-F-14256. For soldering electrical connections, composition SN60 or SN63 shall be used unless another composition is required for a specific application.

3.10.3 Rosin-base residues. - Flux residues shall be removed by applying organic solvents. Mechanical means such as agitation, brushing, etc. may be

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used in conjunction with the solvents. The cleaning solvents and methods used shall not have any deleterious effect on the parts, materials, and assemblies being cleaned.

3.11 Welding. - Wherever practicable, welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858. Spot or other intermittent welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. When spot or other intermittent welds are used to hold a part, the number of welds shall be at least two.

3.11.1 Cleaning prior to welding. - Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.11.2 Process. - Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium, or stainless steel.

3.11.3 Cleaning after welding. - Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed from the part and the surrounding area.

3.12 Wiring and cabling. - Wiring and cabling shall be neat and sturdy.

3.12.1 Cabling. - Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by tape or twine conforming to Specification MIL-T-713, tape conforming to Specification MIL-I-7798; or straps for cable harnessing conforming to Specification MIL-S-23190; however, cord conforming to Specification MIL-I-3158 shall be used where resistance to heat or chemicals is essential. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.12.2 Slack. - Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

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(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected, except for certain radio frequency leads which must be as short as possible for electrical reasons.

(d) To ensure freedom of motion of lugs or terminals normally intended to have some degree of movement (for example, floating contacts on electron tube sockets).

(e) To facilitate field repair of broken or cut wires.

(f) To prevent chafing or breaking of individual wires due to repeated flexing of hinged parts.

3.12.3 Protection. - Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes in metal partitions, shields, or similar items less than 1/8 inch in thickness, the wires shall be protected by suitable grommets or bushings. Where wires run through holes in metal 1/8 inch in thickness or greater, the wires shall be protected by grommets or bushings or shall have the edges of the holes rounded to a minimum radius of 1/16 inch.

3.12.4 Support. - Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable during and after subjection of the equipment to specified service conditions, or after service or repair of the equipment in a normal manner. (Note: When shielding on wire or cable is unprotected by an outer insulation, adequate support is necessary to prevent the shielding from coming in contact with exposed terminals or conductors.) Twine or tape shall not be used for securing wire and cable.

3.12.5 Clearance. - Clearance between wires or cables, and parts such as electron tubes, and resistors, shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.12.6 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.12.7 Insulating sleeving. - Insulating sleeving shall not be used unless specified for a particular application or specifically approved by the contracting officer. Approval will not be granted when other means of insulation are

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practicable. If insulating sleeving is used, it shall conform to Specifications MIL-I-631, MIL-I-3190, MIL-I-7444, MIL-I-18057, MIL-I-22076, MIL-I-22129, MIL-I-21557, or MIL-I-23053.

3.12.8 Inductive and capacitive effects. - Wires and cables shall be so located that inductive and capacitive effects, unless used as a design feature, will be the minimum practicable.

3.12.9 Connections. -

3.12.9.1 General. - Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver-plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and a nonmetallic material. Welded electrical connections shall not be used where it may be necessary to disconnect and reconnect the circuit during servicing. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173 in accordance with Specification MIL-T-152. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections, except that these connections need not be masked during overall treatment with transparent protective coating materials.

3.12.9.2 Part terminations and lug terminals. - Part terminations and lug terminals shall conform to requirement 19 of Standard MIL-STD-454. Terminals conforming to Specification MIL-T-15659 shall be of the types shown on Standards MS 35436 and MS 35440, and terminals conforming to Specification MIL-T-55156 shall be limited to those in Standard MIL-STD-1277.

3.12.10 Grounding. - Ground connection to shields and to other mechanical parts, except the chassis or frame, shall not be made to complete electrical circuits but only to eliminate high potential ac points. Intermediate-frequency transformers, however, may be grounded through their cases if other means of grounding are impracticable.

3.12.10.1 Grounding to chassis. - Grounding to chassis shall conform to Drawing SC-B-61578.

3.12.10.2 Wire and cable shielding. - Wire or cable shielding shall be grounded to the chassis or frame of the equipment. When the wire or cable terminates in a connector, the shielding shall be connected through a pin of the connector, unless it is a coaxial type or another type where this procedure is impractical. Shielding shall end at a sufficient distance from exposed conductors to insure against shorting or arcing between the conductor and the shielding. Where practicable, shielding shall be terminated by a crimp-type ferrule or by soldering at the end of the shielding.

3.13 Plastic material and parts. - Where not machined, plastic material and parts shall have the original smooth or polished surface. Surfaces that have been

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sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

3.14 Test Set, Transponder TS-1809( )/APM-123(V). - (See 4.6.)

3.14.1 Operating controls. - All operating controls shall perform their intended function and shall be sufficiently rugged to withstand repeated operation by unskilled personnel. (See 4.6.1.)

3.14.2 Code selector switch. - The code selector switch shall be capable of providing satisfactory operation after being subjected to the cycling test of 4.6.2.

3.14.3 Power switch. - A three position, center off, toggle switch shall permit Test Set, Transponder TS-1809( )/APM-123(V) to be energized by either a 28 Vdc or 115 Vac power source. When in the 28 Vdc and the 115 Vac positions, this switch shall also complete the respective power circuits to the Mode 4 connector for use by the external Mode 4 equipment. (See 4.6.3.)

3.14.4 Power indicator. - The red power indicator shall illuminate when the test set is energized by the 115 Vac and 28 Vdc sources. (See 4.6.4.)

3.14.4.1 Assembly. - The power indicator assembly shall be water and vapor tight. (See 4.6.4.1.)

3.14.5 Panel lights. - The panel lights shall be capable of providing illumination for nighttime operation of the test set. (See 4.6.5.)

3.14.5.1 Assemblies. - The panel light assemblies shall be water and vapor tight. (See 4.6.5.1.)

3.14.6 Accept and reject indicators. - The white accept and red reject indications shall be easily seen up to a distance of ten feet while the test set is operating in direct sunlight or in the dark. (See 4.6.6.)

3.14.6.1 Assemblies. - The accept and reject indicator assemblies shall be water and vapor tight. (See 4.6.6.1.)

3.14.7 Antenna. - The adjustable antenna locking device shall permit orientation and locking of the antenna in various positions in both of its axes. (See 4.6.7.)

3.14.8 Mode 4 direct-radiation switch. - The Mode 4 direct-radiation switch shall be active in Mode 4 only. (See 4.6.8.)

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3.14.9 Mode 4 verification bits switches. - (See 4.6.9.)

3.14.9.1 Switch marked 1 and OFF. - This switch shall provide test set grounds to the Mode 4 connector. (See 4.6.9.1.)

3.14.9.2 Switch marked 2 and OFF. - This switch shall provide test set grounds to the Mode 4 connector. (See 4.6.9.2.)

3.14.10 Mode 4 code switch. - The Mode 4 code switch shall receive and switch external grounds to the Mode 4 connector. (See 4.6.10.)

3.14.11 Potentiometers. - The equipment shall contain potentiometers and adjustable slugs that are set at approximately the middle of their range and shall be capable of performing the following functions. Meeting the test of 4.6.11 shall be considered as compliance with this requirement.

Adjustment	Function
A1R1	Adjust position of third Mode 4 reply video clock pulse.
A1R2	Adjust position of second Mode 4 reply video clock pulse.
A1R8	Adjust width of third Mode 4 reply video clock pulse.
A1R11	Adjust width of second Mode 4 reply video clock pulse.
A1R24	Adjust width of first Mode 4 reply video clock pulse.
A1R17	Adjust width of Mode 4 reply video line drive generator.
A1R22	Adjust period of Mode 4 video enable delay.
A1R32	Adjust width of challenge video waveform generator.
A1R43	Adjust Mode 4 receiver sensitivity.
A1R29	Adjust position of Mode 4 ISLS trigger.
A2R2	Adjust period of time decoded video gate delay.
A2R7	Adjust width of time decoded video gate.
A3R3	Adjust current sensing limit of +5 Vdc switching regulator.
A3R24	Adjust + 5 Vdc switching regulator output voltage.
A4R6	Adjust width of delay line driver and Mode 4 pretrigger waveforms.
A4R22	Adjust period of ISLS delay.
A4R29	Adjust width of ISLS and Mode 4 interrogation pulse waveforms.

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A4R41	Adjust modulator drive pulse width for interrogation P1 and P3 pulses.
A5R31	Adjust period of PRF generator.
A5R1	Adjust Mode 1 interrogation pulse spacing.
A5R2	Adjust Mode 2 interrogation pulse spacing.
A5R3	Adjust Mode 3/A interrogation pulse spacing.
A5R4	Adjust Mode C interrogation pulse spacing.
A5R5	Adjust TEST Mode interrogation pulse spacing.
A5R6	Adjust P3 position for all interrogation modes.
A7R1	Adjust position of C1 clock pulse.
A7R2	Adjust position of C2 clock pulse.
A7R3	Adjust position of C4 clock pulse.
A7R4	Adjust position of D1 clock pulse.
A7R5	Adjust position of D2 clock pulse.
A7R6	Adjust position of D4 clock pulse.
A7R7	Adjust position of A1 clock pulse.
A7R8	Adjust position of A2 clock pulse.
A7R9	Adjust position of A4 clock pulse.
A7R10	Adjust position of B1 clock pulse.
A7R11	Adjust position of B2 clock pulse.
A7R12	Adjust position of B4 clock pulse.
A7R13	Adjust position of F2 clock pulse.
A7R14	Adjust position of SPI clock pulse.
A8R23	Adjust period of Mode 4 video.
A8R5	Adjust receiver sensitivity.
A10R7	Adjust period of video enable delay.
A11R6	Adjust period of video single-shot.
A11R15	Adjust period of clock single-shot.
A11R30	Adjust period of accept-reject single-shot.
A12R19	Adjust input to reply evaluator.
A14R8	Adjust +150 Vdc regulated supply.
A16R1	Adjust +22 Vdc regulated supply.
A16R15	Adjust amplitude of ISLS pulse.
A16C5	38.1481 MHz oscillator tuning.
A16C11	First tripler tuning.
A16C16	Second tripler tuning.
A16C17	Final tripler drive tuning.
A16C2	Power amplifier output tuning.
A16Z1 POWER	Power amplifier output level adjust.
A16Z1 FREQ.	Power amplifier frequency adjust.

3.14.12 Mode 4 connector. - The Mode 4 connector shall be capable of providing the following inputs and outputs for the Mode 4 interface:

(a) Wire 1 - Mode 4 pretrigger output pulse from the test set to satisfy requirements specified for Interrogator Computer KIR-1A/TSEC in DOD AIMS Specification 64-900. (See 4.6.12(a).)

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(b) Wire 2 - Mode 4 challenge video modulation input to the test set to satisfy requirements specified for Interrogator Computer KIR-1A/TSEC in DOD AIMS Specification 64-900. (See 4.6.12(b).)

(c) Wire 3 - Mode 4 reply video output pulse from the test set to satisfy requirements specified for Interrogator Computer KIR-1A/TSEC in DOD AIMS Specification 64-900. (See 4.6.12(c).)

(d) Wire 4 - Mode 4 time decoded video input to the test set to satisfy requirements specified for Interrogator Computer KIR-1A/TSEC in DOD AIMS Specification 64-900. (See 4.6.12(d).)

(e) Wires 5 and 6 - Plus 28 Vdc and ground controlled by the test set power switch. (See 4.6.12(e).)

(f) Wires 7 and 8 - 115 Vac controlled by the test set power switch. (See 4.6.12(f).)

(g) Wires 9 and 10 - Test set grounds provided by the verification bit 1 switch in the 1 and OFF positions respectively. (See 4.6.12(g).)

(h) Wires 11 and 12 - Test set grounds provided by the verification bit 2 switch in the 2 and OFF positions respectively. (See 4.6.12(h).)

(i) Wire 13 - External ground input to the test set code A/B switch common. (See 4.6.12(i).)

(j) Wires 14 and 15 - External ground outputs provided by the test set code A/B switch in the A and B positions respectively. (See 4.6.12(i).)

(k) Wire 16 - External ground input to the test set mode selector switch for code zeroize alarm indicator. (See 4.6.12(j).)

3.15 Combination case. - The combination case shall consist of a bottom case assembly and a detachable Cover, Test Set CW-890( )/APM-123(V). (See 4.7.)

3.15.1 Bottom case assembly. - (See 4.7.1.)

3.15.1.1 Orientation. - The combination case design shall permit installation of Test Set, Transponder TS-1809( )/APM-123(V) in only one position. (See 4.7.1.)

3.15.1.2 Chassis fasteners. - The Test Set, Transponder TS-1809( )/APM-123(V) shall be contained within the case by means of captive screws that permit secure fastening and ease in removal. (See 4.7.1.)

3.15.2 Cover, Test Set CW-890( )/APM-123(V). - (See 4.7.2.)

3.15.2.1 Contents. - The combination case cover shall provide storage of the accessories specified in 6.3 and the test set's operating spares. (See 4.7.2.)

3.15.2.2 Latches. - The combination case cover latches shall be so placed as to permit the cover to be fastened in only one position. (See 4.7.2.)

3.15.2.3 Design. - The cover shall be capable of being affixed to the bottom of the case assembly during radiation tests to provide additional elevation of the test set. (See 4.7.2.)

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3.15.3 Configuration. - The geometric configuration of the combination case shall permit ease in stacking the AN/APM-123( )(V). (See 4.7.3.)

3.15.4 Relief valves. - The relief valves located on the bottom section and cover of the combination case shall be capable of equalizing the internal and external pressures. (See 4.7.4.)

3.16 Altitude Code Conversion Computer. - The computer shall be capable of converting Mode C reply codes for altitude reporting to station heights in 100 foot increments. (See 4.8.)

3.17 Weight. - The maximum weight for all versions of the AN/APM-123( )(V) shall be 50 pounds. (See 4.9.)

3.18 Size. - The maximum overall dimensions of the AN/APM-123( )(V) shall be 12-3/8 inches high, 13-1/4 inches wide, and 15-9/16 inches long. These measurements shall be made with the latches closed, the shoulder strap removed, and with both the carrying handle and shoulder strap fasteners collapsed against the side of the case. (See 4.10.)

3.19 Reliability. - (See 4.11.)

3.19.1 Reliability program. - The contractor shall institute a reliability program in accordance with Standard MIL-STD-785.

3.19.2 Reliability in Mean Time Between Failure (MTBF). - The equipment shall have a mean time between failures of at least 250 hours operating time when tested in accordance with Test Level E of Standard MIL-STD-781 with the exception that the lowest operating temperature shall be -40°C.

3.19.3 Useful operating life. - When tested in accordance with Test Level E of Standard MIL-STD-781 with the exception that the lowest operating temperature shall be -40°C, the equipment shall have a useful operating life, as defined in Standard MIL-STD-781, of 10,000 hours with no more than normal maintenance and replacement of parts. Demonstration that the equipment is capable of satisfying this requirement is not required unless specified in the contract. (See 6.2(j).)

3.20 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.12.) Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

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3.21 Inspection conditions. - All inspections shall be conducted at 85° ±20°F and at prevailing ambient atmospheric pressures and relative humidities, unless otherwise specified for particular inspections. (See 4.13.)

3.22 Primary input power requirements. - The equipment shall meet specified performance requirements from the following external power sources for Modes 1, 2, 3/A, TEST, or C:

(a) AC power, single phase, 115 Vac ±10%, 50 to 420 Hz. (See 4.14.1.)
(b) DC power, 28 Vdc ±3 Vdc, 1.5 amps. (See 4.14.2.)

3.22.1 Mode 4. - For Mode 4 operation, the test set shall be capable of delivering an additional 1.5 amps maximum at 28 Vdc and 0.5 amps maximum at 115 Vac, over what is specified in paragraph 3.22, to the Mode 4 connector. (See 4.14.3.)

3.23 Voltage protection. - Primary fusing of the input power shall protect the test set from ac overvoltages in excess of 128 Vac and from dc input voltages in excess of 35 Vdc. After the voltages are returned within specified limits and the fuses replaced, the test set shall be capable of resuming normal operation. (See 4.15.)

3.23.1 AC-DC protection. - The ac and dc power sources and power inputs shall not be capable of being interconnected. The input power connector shall use separate pins for the ac and dc input power. (See 4.15.1.)

3.24 Warm-up time. - The equipment warm-up time shall not exceed two minutes when operating at room ambient and at extreme operating temperature conditions. (See 4.16.)

3.25 Transmitter frequency. - The test set transmitter output frequency shall be 1030 MHz ±0.02%. (See 4.17.)

3.26 Transmitter output power. - The transmitter output power level at the test set probe jack shall be internally adjustable over the range of -15 to 0 dBm. (See 4.18.)

3.26.1 Stability. - The transmitter output power shall have a stability of ±3 dB when set to any value within the range of -15 to 0 dBm. This stability shall be maintained under the service conditions specified herein. (See 4.18.1.)

3.26.2 Normal power. - The test set shall be capable of having its normal transmitter output power level set at -9 ±1 dBm. (See 4.18.2.)

3.26.3 Duty cycle. - The test set shall be capable of delivering the power level specified in 3.26 and 3.26.2 while operating at a one percent duty cycle. (See 4.18.3.)

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3.27 Interrogation rate. - (See 4.19.)

3.27.1 Modes 1, 2, 3/A, TEST, and C. - The test set transmitter interrogation rate for Modes 1, 2, 3/A, TEST, and C shall be 400 ±60 interrogations per second. (See 4.19.1.)

3.27.2 Mode 4. - The test set shall be capable of providing a trigger pulse having a repetition rate of 400 ±60 pps for Mode 4 interrogations. The Mode 4 pretrigger input pulse shall comply with the requirements specified for Interrogator Computer KIR-1A/TSEC in DOD AIMS Specification 64-900. (See 4.19.2.)

3.27.2.1 External modulation. - The test set transmitter shall be capable of being externally modulated with a Mode 4 interrogation pulse train having the characteristics specified in 3.28.2 and 3.29.2. (See 4.19.2.1.)

3.28 Pulse spacing. - (See 4.20.)

3.28.1 Modes 1, 2, 3/A, TEST, and C. - (See 4.20.1.) The spacing between the two interrogation pulses shall be as follows:

Mode	Spacing
1	3.0 ±0.2 microseconds
2	5.0 ±0.2 microseconds
3/A	8.0 ±0.2 microseconds
TEST	6.5 ±0.2 microseconds
C	21.0 ±0.2 microseconds

3.28.2 Mode 4. - The interrogation pulse spacing shall be as specified for Interrogator Computer KIR-1A/TSEC Mode 4 challenge video in DOD AIMS Specification 64-900. (See 4.20.2.)

3.29 Pulse shape. - (See 4.21.)

3.29.1 Modes 1, 2, 3/A, TEST, and C. - (See 4.21.1.) The interrogation pulse shapes shall be as follows:

Width	0.8 ±0.1 microseconds at the 50% amplitude points.
Rise Time	Between 0.05 and 0.1 microseconds when measured between the 10% and the 90% points.
Decay Time	Between 0.05 and 0.2 microseconds when measured between the 10% and 90% points.
Amplitude Variation	Not greater than 10% between the various mode interrogation pulses.

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3.29.2 Mode 4. - (See 4.21.2.) The interrogation pulse shapes shall be as follows:

Width	From 0.3 to 0.7 microseconds at the 50% amplitude points.
Rise Time	Between 0.05 and 0.1 microseconds when measured between the 10% and 90% points.
Decay Time	Between 0.05 and 0.2 microseconds when measured between the 10% and 90% points.
Amplitude Variation	Not greater than 10% between any two Mode 4 pulses when the test set transmitter is externally modulated with Mode 4 pulses of equal amplitude.

3.30 Sidelobe suppression. - (See 4.22.)

3.30.1 Amplitude. - The amplitude of the interrogation control pulse shall be equal to the amplitude of the interrogation pulses when the test set ISLS switch is in the "ON" position. With the switch in the "OFF" position, the amplitude of the control pulse shall be capable of being internally adjusted from a level equal to the amplitude of the interrogation pulses to a point at least 10 dB below this level. These amplitude characteristics shall apply to the control pulse that is introduced into the various modes including Mode 4. (See 4.22.1.1 and 4.22.1.2.)

3.30.2 Spacing. - (See 4.22.2.)

3.30.2.1 Modes 1, 2, 3/A, TEST, and C. - The leading edge of the interrogation control pulse in Modes 1, 2, 3/A, TEST, and C shall be spaced 2.0 ±0.15 microseconds from the leading edge of the first interrogation pulse. (See 4.22.2.1.)

3.30.2.2 Mode 4. - The leading edge of the interrogation control pulse in Mode 4 shall be located as specified in DOD AIMS Specification 65-1000, Technical Standard for the ATCRBS/IFF/Mark XII System. (See 4.22.2.2.)

3.30.3 Shape. - The shape of the interrogation control pulse, that is introduced into the various modes including Mode 4, shall conform to the characteristics specified in 3.29.1 through 3.29.2. (See 4.22.3.1 and 4.22.3.2.)

3.31 Receiver bandwidth. - The test set receiver bandwidth shall be 6.5 ±1 MHz when measured at the -3 dB points on the selectivity curve. (See 4.23.)

3.32 Receiver frequency. - The test set shall be capable of having its receiver center frequency set at 1090 MHz ±0.5 MHz. (See 4.24.)

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3.33 Receiver frequency stability. - The test set receiver center frequency shall have a stability of ±1.5 MHz when set in accordance with 3.32. This stability shall be maintained under the service conditions specified herein. (See 4.25.)

3.34 Receiver sensitivity. - The receiver sensitivity level at the test set probe jack shall be capable of being internally adjusted over the range of -15 to 0 dBm. (See 4.26.1 and 4.26.2.)

3.34.1 Stability. - The receiver sensitivity shall have a stability of ±3 dB when set to any value within the range of -15 to 0 dBm. This stability shall be maintained under the service conditions specified herein. (See 4.26.3.)

3.34.2 Normal sensitivity. - The test set shall be capable of having its normal receiver sensitivity level set at -9 ±1 dBm. (See 4.26.4.)

3.35 Video enable delay gate. - For Modes 1, 2, 3/A, TEST, and C, the test set receiver video enable delay gate shall be capable of rejecting all received signals for a period of 1.8 ±0.5 microseconds after the transmission of the last interrogation pulse. (See 4.27.)

3.36 Video enable gate. - For Modes 1, 2, 3/A, TEST, and C, the test set receiver video enable gate shall be capable of opening at the end of the enable delay gate and closing within 6.0 microseconds after the last expected reply pulse. (See 4.28.)

3.37 Interaction. - The Mark X (SIF) mode of operation shall be rendered inoperative during Mode 4 operation. (See 4.29.)

3.38 Decoding capabilities. - (See 4.30.)

3.38.1 Modes 1, 2, 3/A, TEST, and C. - For Modes 1, 2, 3/A, TEST, and C, the test set shall be capable of detecting the presence, absence, and position of every pulse in every pulse train, as specified in 3.43 through 3.50. (See 4.30.1.)

3.38.2 Mode 4. - For Mode 4, the test set shall be capable of pulse decoding Mode 4 transponder system reply pulses as specified in DOD AIMS Specification 64-900. (See 4.30.2.)

3.39 Framing pulse tolerances. - (See 4.31.) For Modes 1, 2, 3/A, TEST, and C, the test set shall be capable of meeting the following transponder reply framing pulse tolerance requirements:

(a) A variation in framing pulse spacing not exceeding ±0.15 microseconds from the nominal value shall cause an accept indication.

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(b) A variation in framing pulse spacing in excess of ±0.3 microseconds from the nominal value will cause a reject indication.

3.40 Information pulse tolerances. - (See 4.32) For Modes 1, 2, 3/A, TEST, and C, the test set shall be capable of meeting the following transponder reply information pulse tolerance requirements:

(a) A variation in information pulse spacing not exceeding ±0.15 microseconds from the nominal value shall cause an accept indication.

(b) A variation in information pulse spacing in excess of ±0.3 microseconds from the nominal value shall cause a reject indication.

3.41 Mode 4 reply pulse tolerances. - (See 4.33.) The test set shall be capable of meeting the following transponder reply pulse tolerance requirements:

(a) A variation in reply pulse spacing not exceeding ±0.15 microseconds from the nominal value shall cause an accept indication.

(b) A variation in reply pulse spacing in excess of ±0.3 microseconds from the nominal value shall cause a reject indication.

3.42 Reply pulse shape. - (See 4.34.)

3.42.1 Modes 1, 2, 3/A, TEST, and C. - (See 4.34.1.) For Modes 1, 2, 3/A, TEST, and C, the test set shall be capable of decoding transponder replies consisting of pulses having the following RF characteristics:

Width	0.45 ±0.1 microseconds at 50% amplitude.
Fall Time	Between 0.05 microseconds and 0.2 microseconds when measured between the 10% and 90% points.
Rise Time	Between 0.05 microseconds and 0.1 microseconds when measured between the 10% and 90% points.
Amplitude Variation	10% or less between all pulses.

3.42.2 Mode 4. - For Mode 4, the test set shall be capable of receiving and decoding Mode 4 transponder system replies as defined in DOD AIMS Specification 64-900. (See 4.34.2.)

3.43 Selective Identification Features (SIF) Mode 1 structure. - For each Mode 1 interrogation from the test set transmitter, the test set shall be capable of decoding a transponder pulse train consisting of two framing pulses and information pulses ranging from zero to a maximum of five. The pulse train to be decoded shall have framing pulses and information pulses nominally spaced at 20.3 and 2.9 microseconds apart respectively. The information pulses shall be measured in 2.9 microsecond multiples from the initial framing pulse; however, the sixth information pulse position (17.4 microseconds) shall not be used. (See 4.35.)

3.44 SIF Mode 2 structure. - For each Mode 2 interrogation from the test set transmitter, the test set shall be capable of decoding a transponder pulse train consisting of two framing pulses and information pulses ranging from zero to a

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maximum of twelve. The pulse train to be decoded shall have framing pulses and information pulses nominally spaced 20.3 and 1.45 microseconds apart respectively. The information pulses shall be measured in 1.45 microsecond multiples from the initial framing pulse, however, the seventh information pulse position (10.15 microseconds) shall not be used. (See 4.36.)

3.45 SIF Mode 3/A structure. - For each Mode 3/A interrogation from the test set transmitter, the test set shall be capable of decoding a transponder pulse train consisting of two framing pulses and information pulses ranging from zero to a maximum of twelve. The pulse train to be decoded shall have framing pulses and information pulses nominally spaced 20.3 and 1.45 microseconds apart respectively. The information pulses shall be measured in 1.45 microsecond multiples from the initial framing pulse, however, the seventh information pulse position (10.15 microseconds) shall not be used. (See 4.37.)

3.46 Mode 4 structure. - For each Mode 4 interrogation from the test set transmitter, the test set shall be capable of receiving and decoding an RF pulse train having characteristics as specified for Transponder Computer KIT-1A/TSEC in DOD AIMS Specification 64-900. For each correct transponder reply train, the test set shall generate a decoded reply video output pulse, also specified in DOD AIMS Specification 64-900 for Interrogator Computer KIR-1A/TSEC, which shall be made available at the Mode 4 connector for use by the external Mode 4 equipment. The test set shall also be capable of receiving time decoded video signals from the Interrogator Computer KIR-1A/TSEC and evaluating these signals in accordance with the Mode 4 reply evaluator requirements of 3.52 through 3.52.2. (See 4.38.)

3.47 Mode C structure. - For each Mode C interrogation from the test set transmitter, the test set shall be capable of decoding a transponder reply pulse train consisting of two framing pulses and information pulses ranging from zero to a maximum of twelve. The transponder pulse train to be decoded shall have information pulses that are in selected combinations which contain the Special Position Identification (SPI) pulse plus the information pulses that are in a normal Mode 2 reply (See 3.44.) with the exception that the D1 pulse is omitted. The SPI pulse is used in conjunction with the D4 pulse only and is spaced 4.35 ±0.1 microseconds after the second framing pulse. The spacing of the remaining pulses are identical to the pulse spacing in a Mode 2 SIF reply. (See 4.39.)

3.48 TEST Mode structure. - For each TEST mode interrogation, the equipment shall be capable of decoding a transponder test mode reply that is identical to a Mode 3/A reply as defined in 3.45. (See 4.40.)

3.49 Identification of Position (I/P) operation. - (See 4.41.) The test set shall be capable of decoding I/P replies having the following code structures when transponder under test is interrogated in Modes 1, 2, and 3/A:

(a) The Mode 1 reply from the transponder shall consist of two pulse trains of the selected code spaced 4.35 ±0.1 microseconds between trains.

(b) The Mode 2 and Mode 3/A replies from the transponder shall consist of a single pulse train of the selected code followed by a single identification pulse spaced 4.35 ±0.1 microseconds after the last framing pulse.

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3.50 Emergency operation. - (See 4.42.) The test set shall be capable of decoding the following coded emergency replies received from the transponder under test:

(a) For Modes 1 and 2, the emergency replies shall consist of four pulse trains spaced 4.35 ±0.1 microseconds apart. The first pulse train shall consist of the selected code and the next three, the framing pulses only.

(b) For Mode 3/A, the emergency replies shall consist of four pulse trains spaced 4.35 ±0.1 microseconds apart. The first pulse train shall consist of code 77(00) and the next three, the framing pulse only.

3.51 Reply rate evaluator. - In Modes 1, 2, 3/A, TEST, and C, the test set shall be capable of providing an accept indication when the transponder under test gives correct replies to 80% or more of the test set's interrogations. The test set shall indicate a reject if the transponder gives correct replies to 50% or less of the test set's interrogations. (See 4.43.)

3.52 Mode 4 reply evaluator. - When testing in Mode 4, the equipment shall be capable of evaluating the replies to 64 of the test set's interrogations and provide an accept or reject indication. (See 4.44.)

3.52.1 Direct test method. - When the direct connection method is used, the test set shall provide an accept indication when 16 or more correct replies in a row are received for 64 interrogations. Receiving less than 16 correct replies in a row shall cause the test set to provide a reject indication. More than one reply or no reply to an interrogation will count as an incorrect reply. (See 4.44.)

3.52.2 Radiation test method. - When the radiation method is used, the test set shall provide an accept indication when any 56 or more correct replies are received for 64 interrogations. Receiving less than 56 correct replies shall cause the test set to provide a reject indication. More than one reply or no reply to an interrogation will count as an incorrect reply. (See 4.44.)

3.53 System test requirement. - (See 4.45.)

3.53.1 Modes 1, 2, 3/A, TEST, and C (See 4.45.1.). - The test set shall be capable of indicating an accept if all of the following transponder system parameters are present:

(a) Correct receiver frequency
(b) Sufficient receiver sensitivity
(c) Correct decoding
(d) Correct coding
(e) Correct transmitter frequency
(f) Sufficient power output
(g) Sufficient number of correct replies

These parameters must all be within acceptable performance criteria reflected in this specification to achieve an accept condition. Any one of the above characteristics failing to meet this criteria shall cause a reject condition.

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3.53.2 Mode 4. - The system test requirements for Modes 1, 2, 3/A, TEST, and C (See 3.53.1.) shall also be applicable for Mode 4 when the test set is used in conjunction with external Mode 4 equipment. (See 4.45.2.)

3.54 Frequency power test requirement. - (See 4.46.)

3.54.1 Modes 1, 2, 3/A, TEST, and C. - (See 4.46.1.) The test set shall be capable of indicating an accept if all of the following transponder system parameters are present:

(a) Correct receiver frequency
(b) Sufficient receiver sensitivity
(c) Correct decoding
(d) Correct transmitter frequency
(e) Sufficient power output
(f) Sufficient number of replies

These parameters must all be within acceptable performance criteria reflected in this specification to achieve an accept condition. Anyone of the above characteristics failing to meet this criteria shall cause a reject condition.

3.54.2 Mode 4. - The test set, when used in conjunction with external Mode 4 equipment, shall be capable of indicating an accept if all of the transponder system parameters specified 3.54.1 are present. (See 4.46.2.)

3.55 Self test. - With the equipment constructed in accordance with the drawings of paragraph 3.3, the self test feature shall be capable of indicating the test set's operational readiness. (See 4.47.)

3.56 Free space coupling. - The equipment shall be capable of interrogating a transponder and evaluating its replies when the test set is resting on the ground and located fifty feet from the transponder antenna which is situated two feet above the ground. The transponder under test shall have a minimum receiver sensitivity of -69 dBm and a minimum output power level of +51 dBm measured at the input to the transponder antenna. (See 4.48.)

3.57 Umbilical coupling. - The equipment shall be capable of interrogating a transponder and evaluating its replies when the test set is coupled to the transponder through a 25 foot umbilical cable and a 50 dB attenuator attached to the antenna connector on the transponder. (See 4.49.)

3.57.1 Antenna AT-884( )/APX-44. - The equipment shall be capable of interrogating a transponder and evaluating its replies when the test set is coupled to the transponder through a 25 foot umbilical cable, a 50 dB attenuator, and a test hood that is attached over the transponder Antenna AT-884( )/APX-44. (See 4.49.1.)

3.57.2 Antenna AT-741( )/A. - The equipment shall be capable of interrogating a transponder and evaluating its replies when the test set is coupled to the transponder through a 25 foot umbilical cable and a test probe located at the AT-741( )/A antenna. (See 4.49.2.)

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3.58 Radio interference suppression. - The equipment shall meet the emanation and susceptibility limits of Specification MIL-I-11748 for class I with the exception that the highest frequency of test shall not exceed 10 GHz. (See 4.50.) Application of suppression components and techniques shall conform to that specification. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance with the specified emanation and susceptibility limits, he shall obtain approval from the contracting officer before proceeding with the modification.

3.59 Operation. - The AN/APM-123( )(V) shall be tested immediately prior to being packaged to demonstrate that it is operational. (See 4.51.)

3.60 Service conditions. - The equipment shall meet the following service conditions. Where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.60.1 Relative humidity. - Exposure to 97 percent relative humidity for 20 hours and 100 percent relative humidity (with condensation) for 4 hours for each 24 hour period. Meeting the test of 4.52 shall be considered compliance with this requirement.

3.60.2 Temperature. -

(a) Operating. - Ambient temperature in the range of +131°F to -40°F for continuous operation and +160°F for intermittent operation. The equipment shall be capable of meeting the test of paragraph 4.53.

(b) Nonoperating. - Exposure in the range of +185°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 12 hours, at any one time. (See 4.53.)

3.60.3 Elevation. -

(a) Operating. - Up to 10,000 feet above sea level. (See 4.54.)

(b) Nonoperating. - Up to 50,000 feet above sea level. (See 4.54.)

3.60.4 Immersion. - Three feet of water for 2 hours. (See 4.55.) This test is to be performed after the test item has been subjected to and passed the shock tests of 4.58 and 4.59 and the bounce test of 4.60.

3.60.5 Salt spray. - The equipment, including case cover, assemblies, and parts, shall be exposed to 48 hours of salt spray in accordance with paragraph 4.56; and shall sustain no corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface. In addition, the equipment shall be capable of meeting full specification performance requirements after being subjected to this test.

3.60.6 Vibration. - The equipment shall not suffer mechanical damage or degradation in performance when subjected to the vibration test of 4.57.

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3.60.7 Shock; bench-handling. - The equipment shall suffer no physical damage after being subjected to the test of paragraph 4.58 and shall be capable of meeting the specified performance for the following parameters:

Inspection	Paragraph	Performance
Transmitter power (Normal)	3.26.2	Full Specification Requirements
Receiver sensitivity (Normal)	3.34.2	Full Specification Requirements
Self test	3.55	Full Specification Requirements

3.60.8 Shock, drop. - The equipment shall suffer only minor physical damage after being subjected to the test of 4.59 and shall be capable of meeting full specification requirements for the measurements specified for the relative humidity test in 4.52.3.

3.60.9 Bounce. - The equipment shall suffer no physical damage after being subjected to the test of 4.60 and shall be capable of meeting full specification requirements for the measurements specified for the relative humidity test in 4.52.3.

3.60.10 Fungus. - Equipment and assemblies shall show no evidence of viable fungus or corrosion* when subjected to 28 days of the fungus test specified in 4.61.

* Corrosion is defined as any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.61 Preconditioning. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of paragraph 4.62. (Also see 4.5.)

3.62 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of Standard MIL-STD-454 and the applicable portions of the following paragraphs herein:

Process	Req Para
Cleaning of parts	3.5.1
Cleaning of units	3.5.2
Riveting	3.8
Securing of parts, general	3.9.1
Brazing and soldering	3.10.1 through 3.10.3
Welding	3.11 through 3.11.3
Wiring and cabling	3.12 through 3.12.10.2
Plastic material and parts	3.13

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) First Article - Preproduction inspection (does not include preparation for delivery). (See 4.3.)
(b) Inspection covered by subsidiary documents. (See 4.4.)
(c) Quality conformance inspection.
(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5.)
(2) Quality conformance inspection of preparation for delivery. (See 4.64.)

4.3 First Article - Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspections specified in the subsidiary documents covering the items listed in 4.4, and the inspections specified for Group A, Group B, and Group C (See Tables I, II, and III respectively). The First Article - preproductio environmental inspections shall be performed as specified for sub-group 1 of Table III; other First Article - Preproduction inspections may precede, follow, or be interspersed between the foregoing.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Inspections	Req Para
Finish	3.6
Marking	3.7 through 3.7.3

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the

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Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7.) Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to Group A, Group B, or Group C inspection, except First Article - Preproduction samples, shall be preconditioned after final assembly. (See 3.61.)

4.5.1 Group A Inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table I. Each lot will be subject to verification, utilizing the procedures of Military Standard MIL-STD-105, and the AQL's indicated in Table I. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.51) shall be last. (See 6.5.)

Table I - Group A inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and mechanical	3.62	4.63		
TS-1809( )/APM-123(V)	3.14	4.6	4 DPHU	15 DPHU
Antenna	3.14.7	4.6.7	for the group	
Potentiometers	3.14.11	4.6.11	combined	
AC-DC protection	3.23.1	4.15.1		
Combination case	3.15	4.7		
Bottom case assembly	3.15.1	4.7.1		
Orientation	3.15.1.1	4.7.1	4 DPHU	15 DPHU
Chassis fasteners	3.15.1.2	4.7.1		
Cover, Test Set CW-890( )/APM-123(V)	3.15.2	4.7.2	for the group	
Contents	3.15.2.1	4.7.2		
Latches	3.15.2.2	4.7.2	combined	
Design	3.15.2.3	4.7.2		
Configuration	3.15.3	4.7.3		
Altitude Code Conversion Computer	3.16	4.8	1%	4%
Electrical				
Power switch	3.14.3	4.6.3		
Power indicator	3.14.4	4.6.4		
Panel lights	3.14.5	4.6.5		
Accept-reject indicators	3.14.6	4.6.6		
Mode 4 direct-radiation switch	3.14.8	4.6.8	2.5 DPHU	*
Mode 4 verification bits switches	3.14.9	4.6.9	for the group	
Switch marked 1 and OFF	3.14.9.1	4.6.9.1		
Switch marked 2 and OFF	3.14.9.2	4.6.9.2	combined	
Mode 4 code switch	3.14.10	4.6.10		
Mode 4 connector	3.14.12	4.6.12		
Interaction	3.37	4.29		

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Table I - Group A inspection (cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Primary DC input power requirements	3.22(b)	4.14.2	2.5 DPHU for the group combined	*
Mode 4	3.22.1	4.14.3		
Transmitter frequency	3.25	4.17		
Transmitter normal power	3.26.2	4.18.2		
Transmitter duty cycle	3.26.3	4.18.3		
Interrogation rate	3.27	4.19		
Modes 1, 2, 3/A, TEST, and C	3.27.1	4.19.1	2.5 DPHU	*
Mode 4	3.27.2	4.19.2		
External modulation	3.27.2.1	4.19.2.1		
Pulse spacing	3.28	4.20	for the group	
Modes 1, 2, 3/A, TEST, and C	3.28.1	4.20.1		
Mode 4	3.28.2	4.20.2	combined	
Pulse shape	3.29	4.21		
Modes 1, 2, 3/A, TEST, and C	3.29.1	4.21.1		
Mode 4	3.29.2	4.21.2		
Sidelobe suppression	3.30	4.22		
Amplitude	3.30.1	4.22.1		
Spacing	3.30.2	4.22.2		
Modes 1, 2, 3/A, TEST, and C	3.30.2.1	4.22.2.1		
Mode 4	3.30.2.2	4.22.2.2		
Shape	3.30.3	4.22.3		
Receiver bandwidth	3.31	4.23		
Receiver frequency	3.32	4.24	2.5 DPHU	*
Receiver normal sensitivity	3.34.2	4.26.4	for the group	
Video enable delay gate	3.35	4.27	combined	
Video enable gate	3.36	4.28		
Decoding capabilities	3.38	4.30		
Modes 1, 2, 3/A, TEST, and C	3.38.1	4.30.1		
Mode 4	3.38.2	4.30.2		
Framing pulse tolerances	3.39	4.31		
Information pulse tolerances	3.40	4.32		
Mode 4 reply pulse tolerances	3.41	4.33		
Reply pulse shape	3.42	4.34	2.5 DPHU	*
Modes 1, 2, 3/A, TEST, and C	3.42.1	4.34.1		
Mode 4	3.42.2	4.34.2	for the group	
SIF Mode 1 structure	3.43	4.35		
SIF Mode 2 structure	3.44	4.36	combined	
SIF Mode 3/A structure	3.45	4.37		
Mode 4 structure	3.46	4.38		
Mode C structure	3.47	4.39		
TEST Mode structure	3.48	4.40		
I/P operation	3.49	4.41		
Emergency operation	3.50	4.42		

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Table I - Group A inspection (cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Reply rate evaluator	3.51	4.43		
Mode 4 reply evaluator	3.52	4.44		
Direct test method	3.52.1	4.44	2.5 DPHU	*
Radiation test method	3.52.2	4.44		
System test	3.53	4.45	for the group	
Modes 1, 2, 3/A, TEST, and C	3.53.1	4.45.1		
Mode 4	3.53.2	4.45.2	combined	
Frequency power	3.54	4.46		
Modes 1, 2, 3/A, TEST, and C	3.54.1	4.46.1		
Mode 4	3.54.2	4.46.2		
Self test	3.55	4.47		
Operation	3.59	4.51	0.1%	*

DPHU - Defects per hundred units
% - % defective
* All electrical and operational defects are considered major.

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table II and to the procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Order of inspection within Group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group B inspection

Inspection	Req Para	Insp Para	AQL
Weight	3.17	4.9	6.5%
Size	3.18	4.10	for the group
Interchangeability	3.20	4.12	combined
Voltage protection	3.23	4.15	
Transmitter output power	3.26	4.18	6.5%
Receiver sensitivity	3.34	4.26	
Free space coupling	3.56	4.48	for the group
Umbilical coupling	3.57	4.49	
Antenna AT-884( )/APX-44	3.57.1	4.49.1	combined
Antenna AT-741( )/A	3.57.2	4.49.2	

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4.5.3 Group C inspection. - This inspection shall be as listed in Table III, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection.

Table III - Group C inspection

Inspection	Req Para	Insp Para
Sub-group 1		
Unit A		
Warm-up time	3.24	4.16
Transmitter output power stability	3.26.1	4.18.1
Receiver frequency stability	3.33	4.25
Receiver sensitivity stability	3.34.1	4.26.3
Temperature	3.60.2	4.53
Primary AC input power	3.22(a)	4.14.1
Humidity	3.60.1	4.52
Power indicator assembly	3.14.4.1	4.6.4.1
Panel light assemblies	3.14.5.1	4.6.5.1
Accept-reject indicator assemblies	3.14.6.1	4.6.6.1
Unit B		
Code selector switch	3.14.2	4.6.2
Transmitter output power stability	3.26.1	4.18.1
Receiver frequency stability	3.33	4.25
Receiver sensitivity stability	3.34.1	4.26.3
Radio interference suppression	3.58	4.50
Elevation	3.60.3	4.54
Relief valves	3.15.4	4.7.4
Salt spray	3.60.5	4.56
Unit C		
Transmitter output power stability	3.26.1	4.18.1
Receiver frequency stability	3.33	4.25
Receiver sensitivity stability	3.34.1	4.26.3
Fungus	3.60.10	4.61
Vibration	3.60.6	4.57
Bounce	3.60.9	4.60
Shock, bench-handling	3.60.7	4.58
Shock, drop	3.60.8	4.59
Immersion	3.60.4	4.55

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Table III - Group C inspection (cont'd)

Inspection	Req Para	Insp Para
Sub-group 2		
Unit A		
Warm-up time	3.24	4.16
Transmitter output power stability	3.26.1	4.18.1
Receiver frequency stability	3.33	4.25
Receiver sensitivity stability	3.34.1	4.26.3
Temperature	3.60.2	4.53
Primary AC input power	3.22(a)	4.14.1
Elevation	3.60.3	4.54
Relief valves	3.15.4	4.7.4
Humidity	3.60.1	4.52
Power indicator assembly	3.14.4.1	4.6.4.1
Panel light assemblies	3.14.5.1	4.6.5.1
Accept-reject indicator assemblies	3.14.6.1	4.6.6.1
Unit B		
Code selector switch	3.14.2	4.6.2
Transmitter output power stability	3.26.1	4.18.1
Receiver frequency stability	3.33	4.25
Receiver sensitivity stability	3.34.1	4.26.3
Radio interference suppression	3.58	4.50
Vibration	3.60.6	4.57
Bounce	3.60.9	4.60
Shock, bench-handling	3.60.7	4.58
Shock, drop	3.60.8	4.59
Immersion	3.60.4	4.55
Salt spray	3.60.5	4.56
Sub-group 3		
Unit A		
Code selector switch	3.14.2	4.6.2
Warm-up time	3.24	4.16
Transmitter output power stability	3.26.1	4.18.1
Receiver frequency stability	3.33	4.25
Receiver sensitivity stability	3.34.1	4.26.3
Temperature	3.60.2	4.53
Primary AC input power	3.22(a)	4.14.1
Elevation	3.60.3	4.54

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Table III - Group C inspection (cont'd)

Inspection	Req Para	Insp Para
Relief valves	3.15.4	4.7.4
Humidity	3.60.1	4.52
Power indicator assembly	3.14.4.1	4.6.4.1
Panel light assemblies	3.14.5.1	4.6.5.1
Accept-reject indicator assemblies	3.14.6.1	4.6.6.1
Unit B		
Transmitter output power stability	3.26.1	4.18.1
Receiver frequency stability	3.33	4.25
Receiver sensitivity stability	3.34.1	4.26.3
Radio interference suppression	3.58	4.50
Fungus	3.60.10	4.61
Vibration	3.60.6	4.57
Bounce	3.60.9	4.60
Shock, bench-handling	3.60.7	4.58
Shock, drop	3.60.8	4.59
Immersion	3.60.4	4.55

Note: The environmental inspections specified for each unit shall be performed in the order shown.

4.5.3.1 Sampling for inspection of equipment. - For production runs of approximately one year's duration, the sampling for Group C inspection and the order of testing shall be as follows:

(a) Three units from first lot submitted for acceptance shall be subjected to the tests of sub-group 1.

(b) Two units shall be subjected to the tests of sub-group 2.

(c) Two units shall be subjected to the tests of sub-group 3.

(d) Two units shall be subjected to the tests of sub-group 2.

(e) Three units from last lot submitted for acceptance shall be subjected to the tests of sub-group 1.

The points at which the tests specified in (b), (c), and (d), above shall be performed during the production run, shall be determined by dividing the number of units to be produced between the first and last lots into equal portions. All samples shall be selected at random.

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For those conditions where the above procedure is not feasible or does not satisfactorily monitor the contractor's production, the Group C sampling plan shall be adjusted and specified in the contract. (See 6.2(c).)

4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of conforming Group B and Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

4.6 Test Set, Transponder TS-1809( )/APM-123(V). - (See 3.14.)

4.6.1 Operating controls test. - Satisfactory operation of the controls while performing the various tests specified herein shall constitute compliance with the requirement of 3.14.1.

4.6.2 Code selector switch test. - Each control of the switch shall be cycled from 0 to 7 and back to 0 a minimum of 10,000 times. After cycling, the unit shall be subjected to the decoding test of 4.30.1 to show compliance with the requirement of 3.14.2.

4.6.3 Power switch test. - Satisfactory operation of the power switch while performing the Mode 4 connector tests of 4.6.12(e) and 4.6.12(f), and the primary input power tests of 4.14.1 and 4.14.2 shall constitute compliance with the requirement of 3.14.3.

4.6.4 Power indicator test. - The power indicator shall be tested to demonstrate compliance with the requirement of 3.14.4.

4.6.4.1 Assembly. - The power indicator assembly shall be examined at the conclusion of the humidity test of 4.52 to determine compliance with the requirement of 3.14.4.1.

4.6.5 Panel lights test. - The panel lights shall be tested to demonstrate compliance with the requirement of 3.14.5.

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4.6.5.1 Assemblies. - The panel light assemblies shall be examined at the conclusion of the humidity test of 4.52 to determine compliance with the requirement of 3.14.5.1.

4.6.6 Accept and reject indicator test. - The accept and reject indicators shall be tested to demonstrate compliance with the requirement of 3.14.6.

4.6.6.1 Assemblies. - The accept and reject indicator assemblies shall be examined at the conclusion of the humidity test of 4.52 to determine compliance with the requirement of 3.14.6.1.

4.6.7 Antenna test. - The antenna shall be tested to demonstrate compliance with the requirement of 3.14.7.

4.6.8 Mode 4 direct-radiation switch test. - When performing the reply rate evaluator test of 4.43 and the Mode 4 reply evaluator test of 4.44, the Mode 4 direct-radiation switch shall be tested to determine compliance with the requirement of 3.14.8.

4.6.9 Mode 4 verification bits switch test. - (See 3.14.9.)

4.6.9.1 Switch marked 1 and OFF. - Satisfactory operation of the switch while performing the Mode 4 connector test of 4.6.12(g) shall constitute compliance with the requirement of 3.14.9.1.

4.6.9.2 Switch marked 2 and OFF. - Satisfactory operation of the switch while performing the Mode 4 connector test of 4.6.12(h) shall constitute compliance with the requirement of 3.14.9.2.

4.6.10 Mode 4 code switch test. - Satisfactory operation of the switch while performing the Mode 4 connector test of 4.6.12(i) shall constitute compliance with the requirement of 3.14.10.

4.6.11 Potentiometer test. - The equipment shall be examined against the various schematics to determine that all potentiometers and adjustable slugs have been provided and properly connected. (See 3.14.11.)

4.6.12 Mode 4 connector test. - The following tests shall be used to determine that the various wires of the Mode 4 connector perform the functions specified in 3.14.12:

(a) Wire 1 - Compliance with the requirement of 3.14.12(a) shall be determined while performing the Mode 4 interrogation rate test of 4.19.2.

(b) Wire 2 - Compliance with the requirement of 3.14.12(b) shall be determined while performing the Mode 4 external modulation test of 4.19.2.1.

(c) Wire 3 - Compliance with the requirement of 3.14.12(c) shall be determined while performing the Mode 4 structure test of 4.38.

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(d) Wire 4 - Compliance with the requirement of 3.14.12(d) shall be determined while performing the Mode 4 reply evaluator test of 4.44.

(e) Wires 5 and 6 - Using a voltmeter, test wires 5 and 6 to determine compliance with the requirement of 3.14.12(e).

(f) Wires 7 and 8 - Using a voltmeter, test wires 7 and 8 to determine compliance with the requirement of 3.14.12(f).

(g) Wires 9 and 10 - Using an ohmmeter, test wires 9 and 10 (verification bit switch in the 1 and OFF positions respectively) to determine compliance with the requirement of 3.14.12(g).

(h) Wires 11 and 12 - Using an ohmmeter, test wires 11 and 12 (verification bit switch in the 2 and OFF positions respectively) to determine compliance with the requirement of 3.14.12(h).

(i) Wires 13, 14, and 15 - Using an ohmmeter, test wires 13 and 15 to determine that they are connected when the Mode 4 code switch is in the A position. Repeat the above, testing wires 13 and 14 with the Mode 4 code switch in the B position. The requirements of 3.14.12(i) and 3.14.12(j) shall be satisfied.

(j) Wire 16 - With power applied to the test set and the mode selector switch in the Mode 4 position, the code zeroize alarm indicator shall illuminate when wire 16 is grounded. (See 3.14.12(k).)

4.7 Combination case test. - (See 3.15.)

4.7.1 Bottom case assembly. - The bottom case assembly shall be tested to determine compliance with the requirements of 3.15.1.1 and 3.15.1.2.

4.7.2 Cover, Test Set CW-890( )/APM-123(V). - The combination case cover shall be tested to determine compliance with the requirements of 3.15.2.1 through 3.15.2.3.

4.7.3 Configuration. - The equipment shall be tested to determine compliance with the requirement of 3.15.3.

4.7.4 Relief valves. - Compliance with the requirement of 3.15.4 shall be determined while performing the elevation test of 4.54.

4.8 Altitude Code Conversion Computer test. - The altitude code conversion wheel shall be examined to determine compliance with the requirement of 3.16.

4.9 Weight. - Each version of the AN/APM-123( )(V), consisting of the combination case, test set, operating spares, and all accessories, shall be weighed to determine compliance with the requirement of 3.17.

4.10 Size. - The AN/APM-123( )(V) shall be measured to determine compliance with the requirement of 3.18.

4.11 Reliability test. - (See 3.19.)

4.11.1 Reliability assurance tests. - The equipment shall be subjected to reliability assurance tests in accordance with Test Level E of Standard MIL-STD-781 with the exception that the lowest operating temperature shall be -40°C. Equipments selected for these tests shall have first passed Group A, and B inspections.

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4.11.2 Accept-reject criteria. - The accept-reject criteria for the reliability verification test plan shall be in accordance with Test Plan III of Standard MIL-STD-781. The accept-reject criteria for the reliability sampling test plan shall be in accordance with Test Plan III of Standard MIL-STD-781.

4.11.3 Performance characteristics. - The performance characteristics to be measured shall be detailed in a test procedure that has been prepared and approved in accordance with Standard MIL-STD-781.

4.11.4 Failure criteria. - The requirements of Standard MIL-STD-781 shall be used to determine when a failure has occurred during the test. In addition, whenever performance characteristics fall below the acceptance requirement of 4.11.3, at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless one caused the other parts to fail.

4.11.5 Reliability program. - The reliability program required by 3.19.1 shall first be approved by the Government before testing is begun.

4.12 Inspection for dimensional interchangeability. - The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.20. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of items to be stored within the case.

(c) Dimensions of cavities, when such dimensions affect inspection of items.

(d) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(e) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(f) Size and form of special threads.

4.13 Inspection conditions. - All inspections shall be performed under the conditions specified in 3.21.

4.14 Primary input power test. (See 3.22.)

4.14.1 AC power. - While operating with the power source of 3.22(a), the test set shall be subjected to the tests specified for relative humidity in 4.52.3 to demonstrate compliance with the requirements of 3.22. Each test shall be performed while the test set voltage is set at 103.5 Vac, 115 Vac, and again at 126.5 Vac. When operating at 103.5 Vac, the test set shall be stabilized at -40°F and when operating at 126.5 Vac, the test set shall be stabilized at 131°F.

4.14.2 DC power. - While operating with the power source of 3.22(b) the test

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set shall be subjected to the transmitter power (normal) test (4.18.2), receiver sensitivity (normal) test (4.26.4), and self test (4.47) to demonstrate compliance with the requirements of 3.22. Each test shall be run while the test set voltage is set at 25 Vdc, 28 Vdc, and again at 31 Vdc.

4.14.3 Mode 4. - The equipment shall be tested to show compliance with the requirement of 3.22.1.

4.15 Voltage protection test. - The test set shall be subjected to the overvoltages as specified in 3.23. After the voltages are returned to normal and the fuses replaced, the equipment shall be subjected to the transmitter power (normal) test of 4.18.2, receiver sensitivity (normal) test of 4.26.4, and self test of 4.47 to show compliance with the requirements of 3.23.

4.15.1 AC-DC protection. - The input power connector and the ac and dc power cables shall be examined and tested to show that interconnection of these currents cannot occur. (See 3.23.1.)

4.16 Warm-up time test. - With the equipment at room temperature, apply power and turn the function switch to the self test position. The accept light shall come on within two minutes from the time that the power was applied. Repeat test with the equipment stabilized at -40°F and again at 131°F. (See 3.24 and 4.53.)

4.17 Transmitter frequency test. - Measure the transmitter frequency at the probe output of the test set using equipment capable of accurately determining compliance with the requirement of 3.25.

4.18 Transmitter output power test. - Measure the transmitter output power over its entire range at the probe output of the test set by using a suitable power meter. The test point A4TP5 shall be grounded during this test for cw operation which will provide a 100% duty cycle. (See 3.26.)

4.18.1 Stability. - The transmitter output power stability shall be determined while performing the following tests: relative humidity (4.52), temperature (4.53), elevation (4.54), salt spray (4.56), vibration (4.57), shock, bench-handling (4.58), shock, drop (4.59), and bounce (4.60). (See 3.26.1.)

4.18.2 Normal power. - While performing the test of 4.18, the transmitter normal power shall be set in accordance with paragraph 3.26.2.

4.18.3 Duty cycle. - The requirement of 3.26.3 shall be considered satisfied if the equipment passes the test of 4.18.

4.19 Interrogation rate test. - (See 3.27.)

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4.19.1 Modes 1, 2, 3/A, TEST, and C. - The transmitter interrogation rate for Modes 1, 2, 3/A, TEST, and C shall be determined by measuring the spacing between successive P1 pulses by using a suitable diode detector and oscilloscope. (See 3.27.1.)

4.19.2 Mode 4. - The spacing between successive Mode 4 pretrigger input pulses shall be measured to determine compliance with the requirement of 3.27.2.

4.19.2.1 External modulation. - Adjust the test set for Mode 4 operation. Furnish a Mode 4 interrogation pulse train having the characteristics specified in 3.28.2 and 3.29.2 to the test set Mode 4 connector. The characteristics of the first and last pulse in the pulse train shall be measured at the test set probe connector. (See 3.27.2.1.)

4.20 Pulse spacing test. - (See 3.28.)

4.20.1 Modes 1, 2, 3/A, TEST, and C. - While performing the test of 4.19.1, the spacing between the P1 and P3 pulses shall be measured when operating in every mode except Mode 4. (See 3.28.1.)

4.20.2 Mode 4. - The spacing characteristics of the Mode 4 interrogation pulse train shall be determined during the test of paragraph 4.19.2.1. (See 3.28.2.)

4.21 Pulse shape test. - (See 3.29.)

4.21.1 Modes 1, 2, 3/A, TEST, and C. - While performing the test of 4.19.1, the shape characteristics of the P1 and P3 pulse pair shall be measured when operating in every mode except Mode 4. (See 3.29.1.)

4.21.2 Mode 4. - The shape characteristics of the Mode 4 interrogation pulse train shall be determined during the test of paragraph 4.19.2.1. (See 3.29.2.)

4.22 Sidelobe suppression test. - (See 3.30.)

4.22.1 Amplitude. - (See 3.30.1.)

4.22.1.1 Modes 1, 2, 3/A, TEST, and C. - The ISLS switch shall be turned to the "ON" position and the amplitude of the interrogation pulses and the interrogation control pulse shall be measured using the setup required for the test of 4.19.1 with the ISLS switch in the "OFF" position, vary and measure the amplitude of the control pulse over its entire range when operating in every mode except Mode 4. (See 3.30.1.)

4.22.1.2 Mode 4. - The ISLS switch shall be turned to the "ON" position

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and the amplitude of the interrogation pulses and the interrogation control pulse shall be measured using the setup required for the test of 4.19.2.1. With the ISLS switch in the "OFF" position, vary and measure the amplitude of the control pulse over its entire range when operating in Mode 4. (See 3.30.1.)

4.22.2 Spacing. - (See 3.30.2.)

4.22.2.1 Modes 1, 2, 3/A, TEST, and C. - While performing the test of 4.22.1.1, measure the spacing between the leading edges of the interrogation control pulse and the first interrogation pulse. (See 3.30.2.1.)

4.22.2.2 Mode 4. - The location of the interrogation control pulse in Mode 4 shall be determined while performing the test of 4.22.1.2 to show compliance with the requirement of 3.30.2.2.

4.22.3 Shape. - (See 3.30.3.)

4.22.3.1 Modes 1, 2, 3/A, TEST, and C. - While performing the test of 4.22.1.1, the shape of the interrogation control pulse, in every mode except Mode 4, shall be determined to show compliance with the requirement of 3.30.3.

4.22.3.2 Mode 4. - While performing the test of 4.22.1.2, the shape of the interrogation control pulse shall be determined to show compliance with the requirement of 3.30.3.

4.23 Receiver bandwidth test. - Figure 1 represents an acceptable setup for conducting this test. Set the AN/APM-123( )(V) function switch to the SYSTEM position and the code control, on both the test set and the AN/UPM-98( ), to 0000. The remaining controls shall be set to the positions required to perform this test. The signal generator shall be adjusted for cw operation at 1090 MHz and the input power level shall be adjusted until the test set just indicates accept. When this point is reached, increase the input power level 3 dB. The signal generator frequency shall then be decreased until the test set just indicates reject. This will establish the lower 3 dB point. To establish the upper 3 dB point, increase the frequency until the test set just indicates reject, after passing through the accept range. Measure the bandwidth to determine compliance with the requirement of 3.31.

4.24 Receiver frequency test. - Calculate the center frequency from the arithmetic mean of the upper and lower frequencies from the 3 dB selectivity curve determined in paragraph 4.23. (See 3.32.)

4.25 Receiver frequency stability test. - The receiver frequency stability shall be determined while performing the following tests: relative humidity (4.52), temperature (4.53), elevation (4.54), salt spray (4.56), vibration (4.57), shock, drop (4.59), and bounce (4.60). (See 3.33.)

4.26 Receiver sensitivity test. - (See 3.34.)

4.26.1 Modes 1, 2, 3/A, TEST, and C. - Figure 1 represents an acceptable

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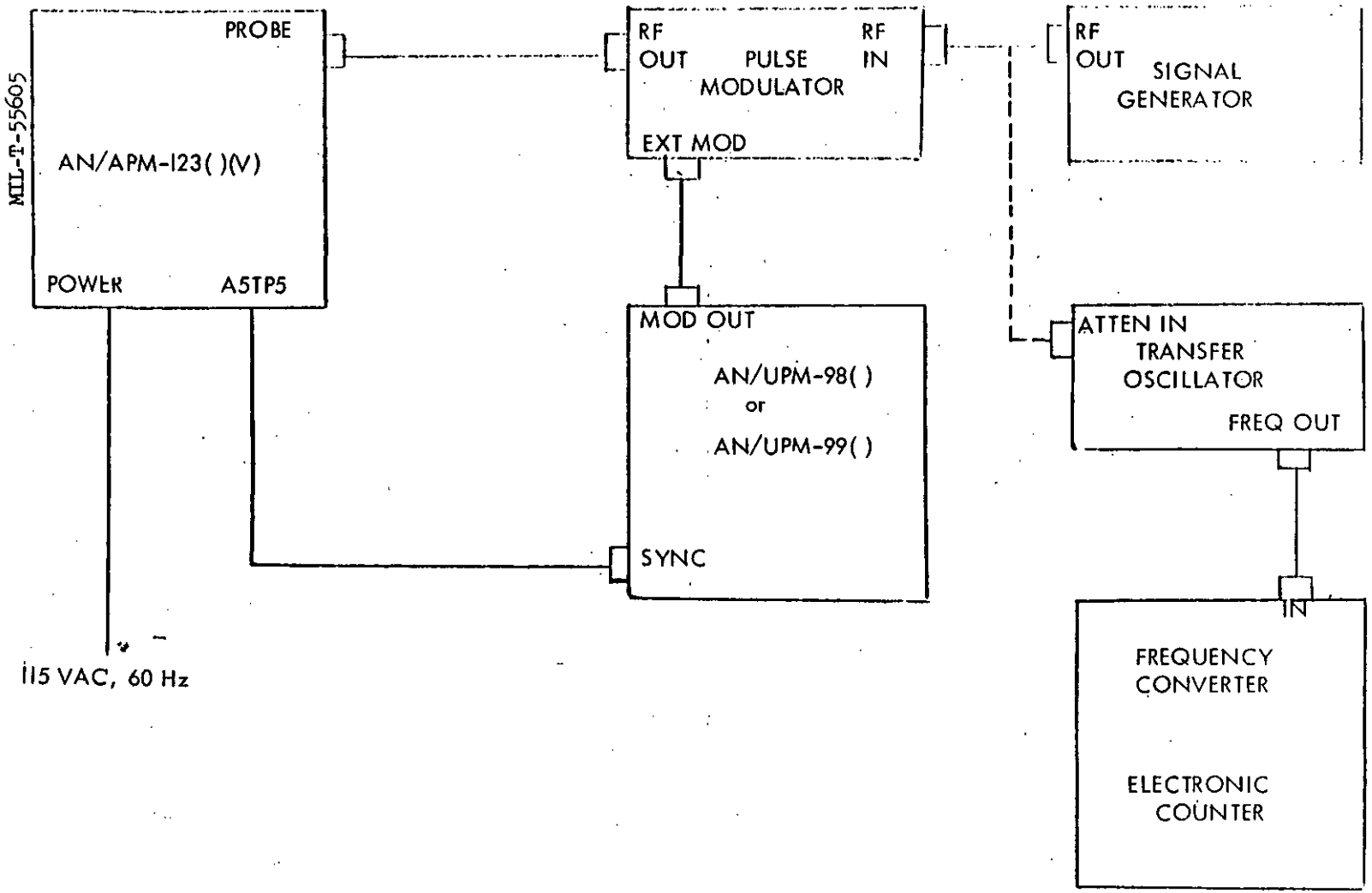


Figure 1. - Receiver Frequency, Sensitivity, and Bandwidth Tests.

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setup for conducting this test. The receiver sensitivity level shall be tested throughout its range to determine compliance with the requirement of paragraph 3.34. This test shall be performed in all modes except Mode 4.

4.26.2 Mode 4. - Figure 2 represents an acceptable setup for conducting this test. The receiver sensitivity level shall be tested throughout its range to determine compliance with the requirement of paragraph 3.34.

4.26.3 Stability. - The receiver sensitivity stability shall be determined while performing the following tests: relative humidity (4.52), temperature (4.53), elevation (4.54), salt spray (4.56), vibration (4.57), shock, bench-handling (4.58), shock, drop (4.59), and bounce (4.60). (See 3.34.1.)

4.26.4 Normal sensitivity. -While performing the test of 4.26.1, set the signal generator frequency to the test set center frequency and set the output power level to -20 dBm. Slowly increase the signal generator power output until the test set just indicates accept. The requirement of paragraph 3.34.2 shall be satisfied.

4.27 Video enable delay gate test. - Figure 3 represents an acceptable setup for conducting this test. Set the AN/APM-123( )(V) function switch to the SYSTEM position and the code control, on both the test set and the AN/UPM-98( ), to 0000. The remaining controls shall be set to the positions required to perform this test. Channel A of the oscilloscope shall be connected to the test set test point A10TP5 and channel B to A5TP3. Measure the spacing from the leading edge of the P3 pulse (Channel B) to the trailing edge of the video enable delay gate (Channel A). This test shall be performed for Modes 1, 2, 3/A, TEST, and C and the requirement of 3.35 shall be satisfied.

4.28 Video enable gate test. - Using the setup of 4.27, with the exception that channel A of the oscilloscope shall be connected to the test set test point A8TP2 and channel B to A9TP4, measure the spacing from the leading edge of the F2 pulse (channel A) to the trailing edge of the enable gate (channel B). This test shall be performed for Modes 1, 2, 3/A, TEST, and C and the requirement of paragraph 3.36 shall be satisfied.

4.29 Interaction test. - A continuity check shall be made at the mode selector switch to determine compliance with the requirement of 3.37.

4.30 Decoding test. - (See 3.38.)

4.30.1 Modes 1, 2, 3/A, TEST, and C. - Figure 3 represents an acceptable setup for conducting this test. Set the AN/APM-123( )(V) function switch to the SYSTEM position and the code control, on both the test set and AN/UPM-98( ), to 0000. The remaining controls shall be set to the positions required to perform this test. The signal generator shall be adjusted for cw operation at 1090 ±1 MHz and the power level to 3 dB above the test set's normal sensitivity level.

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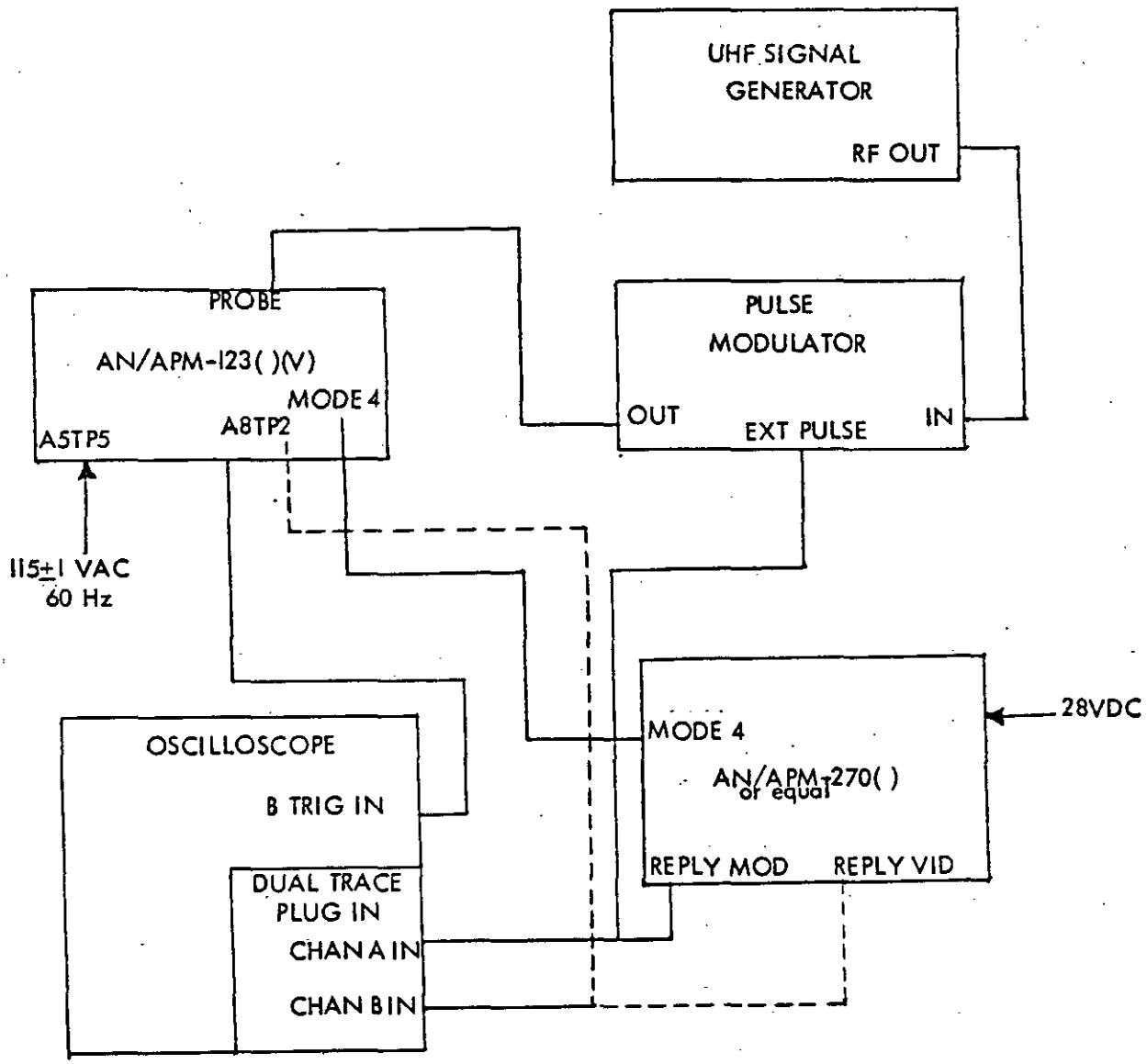


Figure 2. - External video tests.

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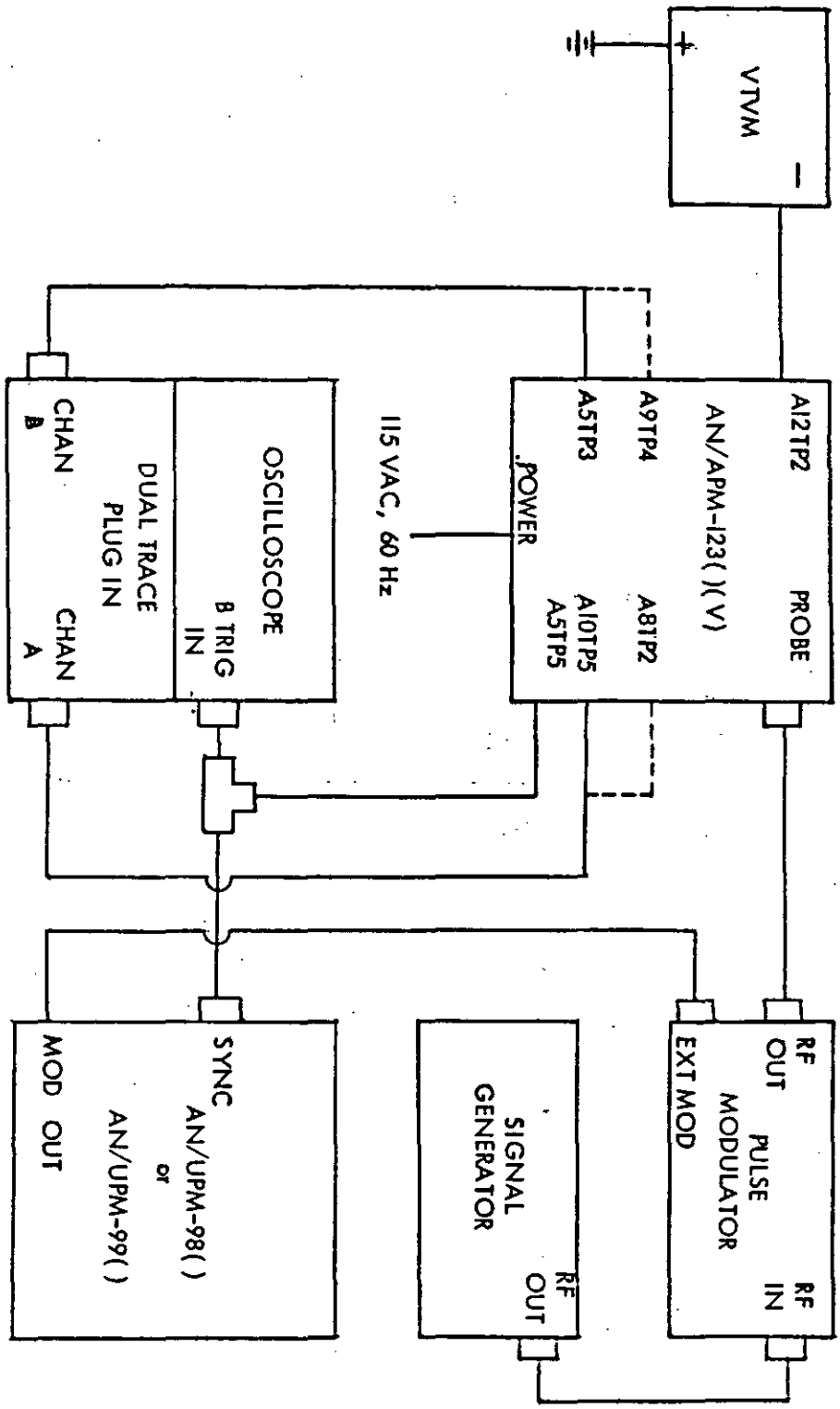


Figure 3. - Gating, Decoding, Framing Pulse, and Information Pulse Tolerance Tests.

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The RF input to the AN/APM-123( )(V) shall have the characteristics specified in paragraph 3.42.1. For Mode C testing only, the signal from the AN/UPM-98( ) shall include the Special Position Identification (SPI) pulse. Under these conditions, the test set shall indicate an accept when operated in each of the modes, except Mode 4.

Repeat this test for each of the following code settings. Both the AN/APM-123( )(V) and the AN/UPM-98( ) shall be set to the same code value. In every mode, except Mode 4, the test set shall indicate an accept when operated with each of the code settings.

Code Selector Position	A	B	C	D
	1	1	1	1
	2	2	2	2
	3	3	3	3
Code	4	4	4	4
	5	5	5	5
Setting	6	6	6	6
	7	7	7	7
	0---→ 7	0	0	0
	0	0---→ 7	0	0
	0	0	0---→ 7	0
	0	0	0	0---→ 7

Note: The arrow (---→) indicates varied through.

When performing this test, the AN/APM-123( )(V) shall indicate a reject as soon as the code setting on either the test set or the AN/UPM-98( ) is changed by one digit. (See 3.38.1.)

4.30.2 Mode 4. - Figure 2 represents an acceptable setup for conducting this test. With the AN/APM-123( )(V) operating in Mode 4, set the signal generator for cw operation at a frequency of 1090 ±1 MHz and set the power level for 3 dB above normal sensitivity. Adjust the AN/APM-270( )(V) or equal to transmit 14, 15, 16, 54, 55, 56, 57, and 64 correct replies, having characteristics as specified in paragraph 3.44.2, to the AN/APM-123( )(V) for testing when in the direct and radiation methods of operation. With the test set switch in the DIR position, the unit shall indicate an accept when receiving 16 and 64 correct replies and a reject for 14 and 15 correct replies. With the test set switch in the RAD position, the unit shall indicate an accept when receiving 56, 57, and 64 correct replies and a reject for 54 and 55 correct replies. (See 3.38.2.)

4.31 Framing pulse tolerance test. - While performing the decoding test of 4.30.1, vary the framing pulse spacing in the AN/UPM-98( ) reply train to demonstrate that the test set complies with the requirements of 3.39.

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4.32 Information pulse tolerance test. - While performing the decoding test of 4.30.1, vary the information pulse spacing in the AN/UPM-98( ) reply train to demonstrate that the test set complies with the requirements of 3.40.

4.33 Mode 4 reply pulse tolerance test. - While performing the decoding test of 4.30.2, vary the reply pulse spacing in the AN/APM-270( ) or equal, reply train to demonstrate that the test set complies with the requirements of 3.41.

4.34 Reply pulse shape test. - (See 3.42.)

4.34.1 Modes 1, 2, 3/A, TEST, and C. - The requirement of 3.42.1 shall be considered satisfied if the equipment passes the tests for receiver normal sensitivity (4.26.4), Modes 1, 2, 3/A, TEST, and C decoding capabilities (4.30.1), framing pulse tolerances (4.31), and information pulse tolerances (4.32).

4.34.2 Mode 4. - The requirement of 3.42.2 shall be considered satisfied if the equipment passes the Mode 4 reply evaluator test of 4.44.

4.35 Selective Identification Features (SIF) Mode 1 structure test. - The requirement of 3.43 shall be considered satisfied if the equipment passes the decoding test of 4.30.1.

4.36 SIF Mode 2 structure test. - The requirement of 3.44 shall be considered satisfied if the equipment passes the decoding test of 4.30.1.

4.37 SIF Mode 3/A structure test. - The requirement of 3.45 shall be considered satisfied if the equipment passes the decoding test of 4.30.1.

4.38 Mode 4 structure test. - The requirement of 3.46 shall be considered satisfied if the equipment passes the Mode 4 decoding test of 4.30.2.

4.39 Mode C structure test. - The requirement of 3.47 shall be considered satisfied if the equipment passes the decoding test of 4.30.1.

4.40 TEST Mode structure test. - The requirement of 3.48 shall be considered satisfied if the equipment passes the decoding test of 4.30.1.

4.41 Identification Position (I/P) operation test. - While performing the decoding test of 4.30.1, change the AN/APM-123( )(V) function switch to the I/P position and adjust the AN/UPM-98( ) to provide an I/P pulse train in Modes 1, 2, and 3/A. The requirement of 3.49 shall be satisfied.

4.42 Emergency operation test. - While performing the decoding test of 4.30.1, change the AN/APM-123( )(V) function switch to the EMER position and adjust the AN/UPM-98( ) to provide an emergency pulse train in Modes 1, 2, and 3/A. The requirement of 3.50 shall be satisfied.

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4.43 Reply rate evaluator test. - The setup for the decoding test of 4.30.1 shall be used to perform this test. With the test set receiving a signal consisting of 100% accepts, calibrate the VTVM so that the voltage indicated is zero. Set the AN/UPM-98( ) to any other code for 0% returns and note the voltage indicated on the VTVM for reference. Adjust the AN/UPM-98( ) substitute pulse position control for 50% returns (50% of reference voltage) and then set the substitute pulse position control for 80% returns (20% of reference voltage). This test shall be performed for Modes 1, 2, 3/A, TEST, and C. The requirements of paragraph 3.51 shall be satisfied.

4.44 Mode 4 reply evaluator test. - While performing the Mode 4 decoding test of 4.30.2, repeat the test with the AN/APM-270( )(V) or equal reply modulation switch set in the 2X position. The AN/APM-123( )(V) shall indicate a reject for all replies received under these conditions. Compliance with the requirements of 3.52 through 3.52.2 shall be considered satisfied if the test set passes this test and that for Mode 4 decoding.

4.45 System test. - (See 3.53.)

4.45.1 Modes 1, 2, 3/A, TEST, and C. - The requirement of 3.53.1 shall be considered satisfied if the equipment has met all those requirements describing parameters pertaining to every mode except Mode 4 that are covered in paragraphs 3.25 through 3.52.

4.45.2 Mode 4. - The requirement of 3.53.2 shall be considered satisfied if the equipment has met all those requirements describing parameters pertaining to Mode 4 that are covered in paragraphs 3.25 through 3.52.

4.46 Frequency power test. - (See 3.54.)

4.46.1 Modes 1, 2, 3/A, TEST, and C. - The requirement of 3.54.1 shall be considered satisfied if the equipment has met all those requirements describing parameters pertaining to every mode except Mode 4 that are covered in paragraphs 3.25 through 3.52.

4.46.2 Mode 4. - The requirement of 3.54.2 shall be considered satisfied if the equipment has met all those requirements describing parameters pertaining to Mode 4 that are covered in paragraphs 3.25 through 3.52.

4.47 Self test. - The test set shall be properly alined prior to start of this test. With the power switch on AC and the function switch on SELF TEST, the TS-1809( )/APM-123(V) shall indicate an accept. The transmitter output power shall then be reduced by approximately 2 dB and the reject light shall come on. Return the transmitter output power to its normal value and the accept light shall now come on. The receiver sensitivity shall then be reduced by approximately 2 dB and the test set shall indicate a reject. Return the

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receiver sensitivity to its normal value and the test set shall now indicate an accept. (See 3.55.)

4.48 Free space coupling test. - Using a suitable receiver-transmitter such as RT-744( )/APX-68 or RT-859( )/APX-72, Antenna AT-884( )/APX-44, and Test Set, Transponder Set AN/APM-239( ) or equal, the AN/APM-123( )(V) shall be tested by setting its function switch to the SYSTEM position and the same code on both the transponder and the test set. Under these conditions, the test set shall indicate an accept; change the code on either unit and the test set shall indicate a reject. (See 3.56.)

4.49 Umbilical coupling test. - (See 3.57.)

4.49.1 Antenna AT-884( )/APX-44. - Using a suitable receiver-transmitter such as RT-744( )/APX-68 or RT-859( )/APX-72, Antenna AT-884( )/APX-44, Test Set, Transponder Set AN/APM-239( ) or equal, Cable Assembly, Radio Frequency CG-409G/U (25' 0"), Attenuator, Fixed CN-1088( )/U, and Test Hood, Antenna MX-4396( )/APM-123(V), the AN/APM-123( )(V) shall be tested by setting its function switch to the SYSTEM position and the same code on both the transponder and the test set. Under these conditions, the test set shall indicate an accept; change the code on either unit and the test set shall indicate a reject. (See 3.57.1.)

4.49.2 Antenna AT-741( )/A. - Using a suitable receiver-transmitter such as RT-744( )/APX-68 or RT-859( )/APX-72, Antenna AT-741( )/A, Test Set, Transponder Set AN/APM-239( ) or equal, Cable Assembly, Radio Frequency CG-409G/U (25' 0"), and a 32 ±1 dB attenuator, the AN/APM-123( )(V) shall be tested by setting its function switch to the SYSTEM position and the same code on both the transponder and the test set. Under these conditions, the test set shall indicate an accept; change the code on either unit and the test set shall indicate a reject. Since the attenuation between the antenna test probe and the transponder shall be 18 ±2 dB, the 32 dB attenuator shall be required for performing this test. This attenuator is not supplied as part of the AN/APM-123( )(V). (See. 3.57.2.)

4.50 Radio interference suppression test. - The AN/APM-123( )(V) shall be tested to show compliance with the requirement of 3.58.

4.51 Operation test. - Power shall be applied to the equipment and the test set function switch shall be placed in the SELF TEST position. The accept light shall come on. (See 3.59.)

4.52 Relative humidity. - (See 3.60.1) The equipment shall be tested as follows:

4.52.1 Test conditions. -

(a) Do not move equipment from the humidity chamber for measurements.

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(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Perform the entire test with the cover removed from the equipment. The cover, however, shall also be subjected to this test.

4.52.2 Test method. -

(a) Dry at 130° ±5° F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.52.3, and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.52.3 shall be made during the periods shown on the standard and shall comply with 4.52.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those controls located on the front panel of the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.52.3.

4.52.3 Performance. - Measurements made during test and specified performance during cycling shall be as follows:

Paragraph	Measurements	Performance during cycling
3.25	Transmitter frequency	Full Specification Requirements
3.26.2	Transmitter power(normal)	Full Specification Requirements
3.27.2.1	Mode 4 external modulation	Full Specification Requirements
3.28 through 3.28.2	Transmitter pulse spacing	Full Specification Requirements
3.29 through 3.29.2	Transmitter pulse shape	Full Specification Requirements
3.30 through 3.30.3	Sidelobe suppression	Full Specification Requirements
3.31	Receiver bandwidth	Full Specification Requirements
3.32	Receiver frequency	Full Specification Requirements
3.34.2	Receiver sensitivity (normal)	Full Specification Requirements
3.38 through 3.38.2	Decoding capabilities	Full Specification Requirements
3.39	Framing pulse tolerances	Full Specification Requirements
3.40	Information pulse tolerances for SIF code 7777 only	Full Specification Requirements
3.42 through 3.42.2	Reply pulse shape	Full Specification Requirements
3.52 through 3.52.2	Mode 4 reply evaluator	Full Specification Requirements

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4.52.4 Visual inspection and failure criteria. - The equipment shall be visually inspected and a record made of any damage or deterioration resulting from the test. When practicable, visual inspection shall be performed within the chamber and at test conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external part which could in any manner prevent the equipment from meeting operational or maintenance requirements shall provide reason to consider the equipment as having failed the test.

4.52.5 Failure. - If any equipment fails to meet the performance specified in 4.52.3 during cycling, it shall be realined or readjusted once and then re-tested. If the equipment then fails to meet 4.52.3 or fails subsequently during cycling, it is considered to have failed the test. In addition, if the equipment fails to meet full specification requirements after the final specified conditioning and adjustment, it is considered to have failed the test.

4.53 Temperature. - (See 3.60.2.) The entire AN/APM-123( )(V) test item, with the cover off, shall be subjected to the temperature cycle shown on Standard MIL-STD-169 with the following exceptions:

(a) Step 2A - Temperature shall be 185°F.
(b) Step 2A - Exposure shall be for 4 hours.
(c) Step 2B - Delete.
(d) Step 3 - Temperature shall be 160°F. Three cycles shall be performed; each cycle shall consist of a 20 minute operation period followed by a 15 minute off period.
(e) Step 4 - Temperature shall be 131°F.
(f) Step 6A - Exposure shall be for 12 hours.
(g) Step 6B - Delete.
(h) Step 7 - Delete.
(i) Step 9 - Delete.

4.53.1 Performance. - Performance during the temperature test shall be as follows:

(a) At steps 1, 4, 5, 8, and 10, the measurements to be made and the required performance shall be as is specified for the relative humidity test in 4.52.3.

(b) At steps 4 and 8, warm-up time (3.24) shall be measured and meet full specification requirements.

(c) At step 3, transmitter power (normal) (3.26.2), receiver sensitivity (normal) (3.34.2), and self test (3.55) shall be measured and meet full specification requirements.

4.54 Elevation test. - (See 3.60.3.) The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the measurements specified for the relative humidity test in 4.52.3.

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(b) Reduce the barometric pressure to 20.6 ±0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.54(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ±0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.54.(a). The equipment shall meet specified performance for these measurements.

4.55 Immersion test. - The equipment, as prepared for field transportation, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ±2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2 hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage. (See 3.60.4.)

4.56 Salt spray test. - The equipment and assemblies shall be exposed in their operating configuration to salt spray in accordance with Method 811.1 of Federal Test Method Standard No. 151. A five percent salt solution shall be used and the length of the test shall be 48 hours. The equipment shall be examined and tested to determine compliance with the requirements of 3.60.5. The measurements to be made at the conclusion of this test and the required performance shall be as is specified for the relative humidity test in 4.52.3.

4.57 Vibration test. - The vibration test shall be performed on the equipment with the cover in place. The equipment shall be mounted using a suitable apparatus to assure that mounting is free from resonances over the test frequency range. The test set shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.03 inch maximum total excursion), the frequency being varied uniformly between the limits of 10 Hz and 55 Hz. The entire frequency range, from 10 to 55 Hz and return to 10 Hz, shall be traversed in one minute. This motion shall be applied for a period of two hours in each of three mutually perpendicular directions (total of 6 hours). To show compliance with the requirement of 3.60.6, the measurements specified for the relative humidity test in 4.52.3 shall be made and satisfied prior to and at the conclusion of this test. At the conclusion of vibration in each of the three axes, the equipment shall also show satisfactory operation in accordance with the self check test of 4.47. In addition, the entire test item shall be inspected for any evidence of mechanical failure.

4.58 Shock test, bench-handling. - The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable

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position for servicing on a solid, 2 inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat 4.58(a), with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

The equipment shall be tested to show compliance with the requirement of 3.60.7 at the conclusion of this test.

4.59 Shock test, drop. - The equipment, in its transit or combination case, shall be dropped a distance of 2 feet on each corner, edge, and face (Total of 26 drops). The drops shall be made from a quick-release hook or from a pendulum tester as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The floor or barrier receiving the impact shall be of 2 inch thick fir backed by concrete or a rigid steel frame. The equipment shall be tested to show compliance with the requirement of 3.60.8 at the conclusion of this test.

4.60 Bounce test; Class F units. - The equipment shall be tested on the package tester, Type 1000-SC, as made by the L. A. B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

(a) Cover the tester bed with a panel of 1/2 inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6 inch intervals around all four edges and at 3 inch intervals in a 6 inch square in the center.

(b) Place the equipment, in its transit case and prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 2 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ±2 rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

The equipment shall be tested to show compliance with the requirement of 3.60.9 at the conclusion of this test.

4.61 Fungus test. - The equipment and assemblies shall be subjected to 28 days of the fungus test specified in Method 508 of Standard MIL-STD-810. For gasket or other non-hermetically sealed equipments, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The equipment shall be subjected to the test in this open condition. (See 3.60.10.)

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4.62 Bounce preconditioning. - The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the package tester, Type 1000-SC, as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection. (See 3.61.)

4.63 Visual and mechanical inspection. - The AN/APM-123( )(V) shall be examined for the defects listed in Standard MIL-STD-252.

4.64 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified. (See 6.2(d).)

5.1.1 Level A. -

5.1.1.1 Cleaning. - Each Test Set, Transponder AN/APM-123( )(V) shall be cleaned in accordance with process C-1 of Specification MIL-P-116.

5.1.1.2 Drying. - Each Test Set, Transponder AN/APM-123( )(V) shall be dried in accordance with the applicable procedure of Specification MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in Specification MIL-P-116 as specified herein:

5.1.1.4.1 Technical literature. - Each technical manual shall be packaged Method 1C-1.

5.1.1.4.2 Running spares, accessories, and ancillary items. - Running spare parts, accessories, and ancillary items, as required, shall be packaged as follows:

5.1.1.4.2.1 Spare fuses. - Spare fuses shall be packaged in accordance with Method III as follows: Cushion the fuses by wrapping together in paperboard conforming to Specification PPP-P-291, Type 1. Secure wrap with gummed paper tape conforming to Specification PPP-T-45, Grade A.

5.1.1.4.2.2 Spare lamps. - Spare lamps shall be packaged in accordance with Method III as follows: Cushion each lamp by wrapping in paperboard conforming to

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PPP-P-291, Type 1. Secure wrap with gummed paper tape conforming to PPP-T-45, Grade A.

5.1.1.4.2.3 Consolidated package. - Consolidate the items packaged as specified in 5.1.1.4.2.1 and 5.1.1.4.2.2 within a cotton bag conforming to PPP-B-20.

5.1.1.4.2.4 Stowing. - Stow the consolidated package within the cover of the test sets. Fill all voids with paperboard conforming to PPP-P-291, Type 1, to prevent movement. Secure inner cover of test set cover with fastenings provided.

5.1.1.4.2.5 Test Set, Transponder AN/APM-123( )(V). - Each test set shall be packaged Method III as follows: Place the test set cover over the test set and secure with spring catches provided. Cushion each test set on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, W6c, designed to absorb the shock of impact in handling and transit. Place the cushioned test set within a close fitting fiberboard box conforming to PPP-B-636, W5c, weather resistant. Place the technical manual packaged as specified in 5.1.1.4.1 in the box directly under the lid of the box. Box closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. - Each Test Set, Transponder AN/APM-123( )(V) shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be level A, B, or C as specified. (See 6.2(d).)

5.2.1 Level A. - A quantity of Test Sets, Transponder AN/APM-123( )(V) packaged as specified in 5.1 shall be packed within a close fitting fiberboard box conforming to Specification MIL-B-43291 or PPP-B-640, class weather resistant, not to exceed the gross weight limitation of 200 pounds. Closure shall be in accordance with the appendix of the applicable box specification. The shipping container shall be reinforced by pressure sensitive filament tape banding conforming to PPP-T-97, Type IV, in accordance with the appendix of the applicable box specification.

5.2.2 Level B. - A quantity of Test Sets, Transponder AN/APM-123( )(V) packaged as specified in 5.1 shall be packed in accordance with 5.2.1.

5.2.3 Level C. - A quantity of Test Sets, Transponder AN/APM-123( )(V) packaged as specified in 5.1 shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

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5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. - Test Set, Transponder AN/APM-123( )(V) is a rugged, portable flight line test equipment which provides a rapid go-no-go evaluation of the operational capabilities of AIMS and US National Standards Transponders. The test set may be operated by relatively unskilled personnel for the performance of free space coupling or umbilical connection tests using the accessories supplied as part of the equipment.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.
(b) Version required.
(c) Group C sampling plan required (See 4.5.3.1.)
(d) Level of packaging and level of packing required for shipment. (level A, level B, or level C) (See Section 5.)
(e) The specific paragraphs of section 5 which are applicable to the particular procurement.
(f) First Article - Preproduction inspection:
Number of sample units required of each item cited in section 1.
(g) Marking and shipping of samples.
(h) Place of final inspection.
(i) Technical literature required.
(j) Useful operating life (See 3.19.3.)

6.3 Classification. - The components that make up the various versions of Test Set, Transponder AN/APM-123( )(V) shall be as follows:

(a)	Test Set, Transponder AN/APM-123( )(V)1	
(1)	Test Set, Transponder	TS-1809( )/APM-123(V)
(2)	Cover, Test Set	CW-890( )/APM-123(V)
(3)	Shoulder Strap, Test Set	MX-7344( )/APM-123(V)
(4)	Test Hood, Antenna	MX-4396( )/APM-123(V)
(5)	Attenuator, Fixed	CN-1088( )/U
(6)	Cable Assembly, Power, Electrical	CX-8330( )/APM-123(V)1
(7)	Cable Assembly, Power, Electrical	CX-8331( )/APM-123(V)1
(8)	Cable Assembly, Radio Frequency	CG-409G/U (25' 0")
(9)	Cable Assembly, Radio Frequency	CG-3308( )/U (6")
(10)	Adapter Connector	UG-564/U
(11)	Cable Assembly, Mode 4	
(12)	Altitude Code Conversion Computer	
(b)	Test Set, Transponder AN/APM-123( )(V)2	
(1)	Test Set, Transponder	TS-1809( )/APM-123(V)

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	(2)	Cover, Test Set	CW-890( )/APM-123(V)
	(3)	Shoulder Strap, Test Set	MX-7344( )/APM-123(V)
	(4)	Test Hood, Antenna	MX-4396( )/APM-123(V)
	(5)	Attenuator, Fixed	CN-1088( )/U
	(6)	Cable Assembly, Power, Electrical	CX-11626( )/APM-123(V)
	(7)	Cable Assembly, Power, Electrical	CX-11627( )/APM-123(V)
	(8)	Cable Assembly, Power, Electrical	CX-11628( )/APM-123(V)2
	(9)	Cable Assembly, Radio Frequency	CG-409G/U (25' 0")
	(10)	Cable Assembly, Radio Frequency	CG-3308( )/U (6")
	(11)	Adapter Connector	UG-564/U
	(12)	Cable Assembly, Mode 4	
	(13)	Altitude Code Conversion Computer	
(c)	Test Set, Transponder AN/APM-123( )(V)3		
	(1)	Test Set, Transponder	TS-1809( )/APM-123(V)
	(2)	Cover, Test Set	CW-890( )/APM-123(V)
	(3)	Shoulder Strap, Test Set	MX-7344( )/APM-123(V)
	(4)	Attenuator, Fixed	CN-1088( )/U
	(5)	Cable Assembly, Power, Electrical	CX-11626( )/APM-123(V)
	(6)	Cable Assembly, Power, Electrical	CX-11627( )/APM-123(V)
	(7)	Cable Assembly, Power, Electrical	CX-11629( )/APM-123(V)3
	(8)	Cable Assembly, Radio Frequency	CG-409G/U (25' 0")
	(9)	Cable Assembly, Radio Frequency	CG-3308( )/U (6")
	(10)	Adapter Connector	UG-564/U
	(11)	Cable Assembly, Mode 4	
	(12)	Altitude Code Conversion Computer	

6.4 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2.)

6.5 Location of operational inspection. - It is desirable that the operational inspection (See 4.51) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example AN/APM-123W(V)1. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.7 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance

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provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodians:
Army - EL
Navy - AS
Air Force - 11

Preparing activity:
Army - EL

Project No. 6625-0458

Reviewer:

Army- AV

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-T-55605 Test Set, Transponder AN/APM-123()(V)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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